C Programming - Platform Questions and Answers

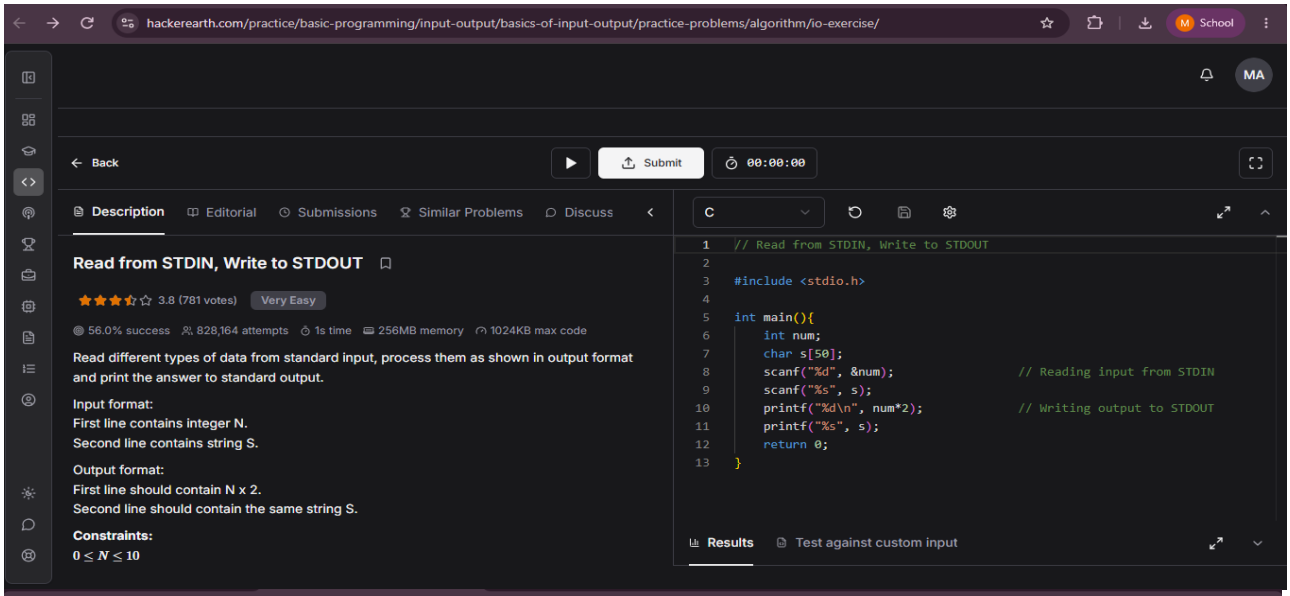
Module 1 — Input/output, variables and basic arithmetic

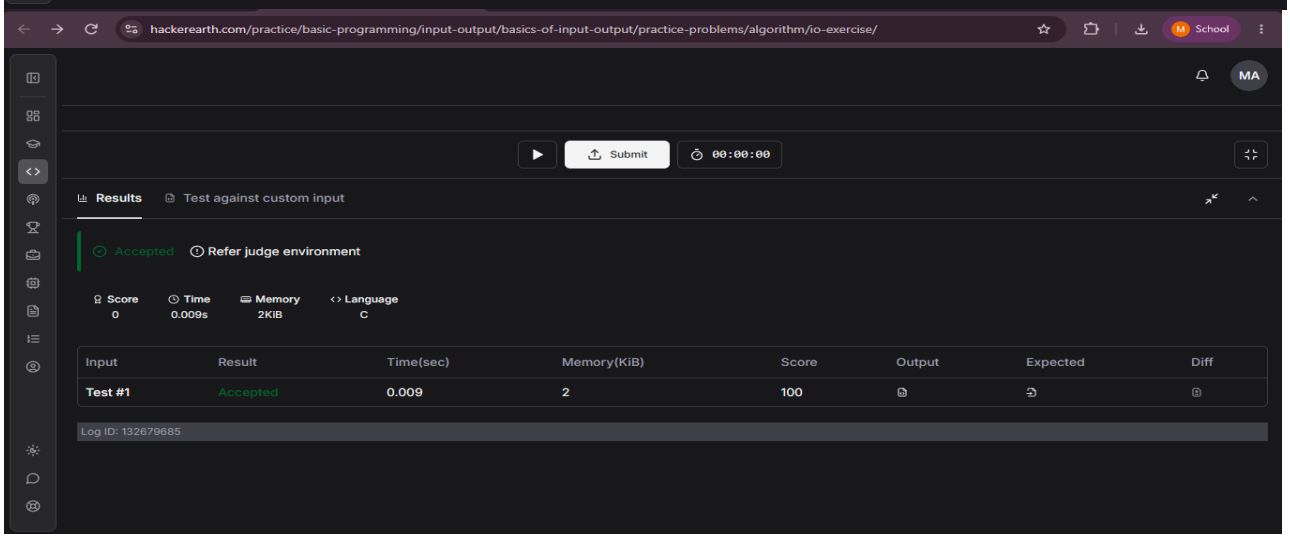
Question 1: Read from STDIN, Write to STDOUT

Answer:

```c
// Read from STDIN, Write to STDOUT
#include <stdio.h>

int main(){
    int num;
    char s[50];
    scanf("%d", &num);                  // Reading input from STDIN
    scanf("%s", s);
    printf("%d\n", num*2);              // Writing output to STDOUT
    printf("%s", s);
    return 0;
}
```





Question 2: Solve Me First

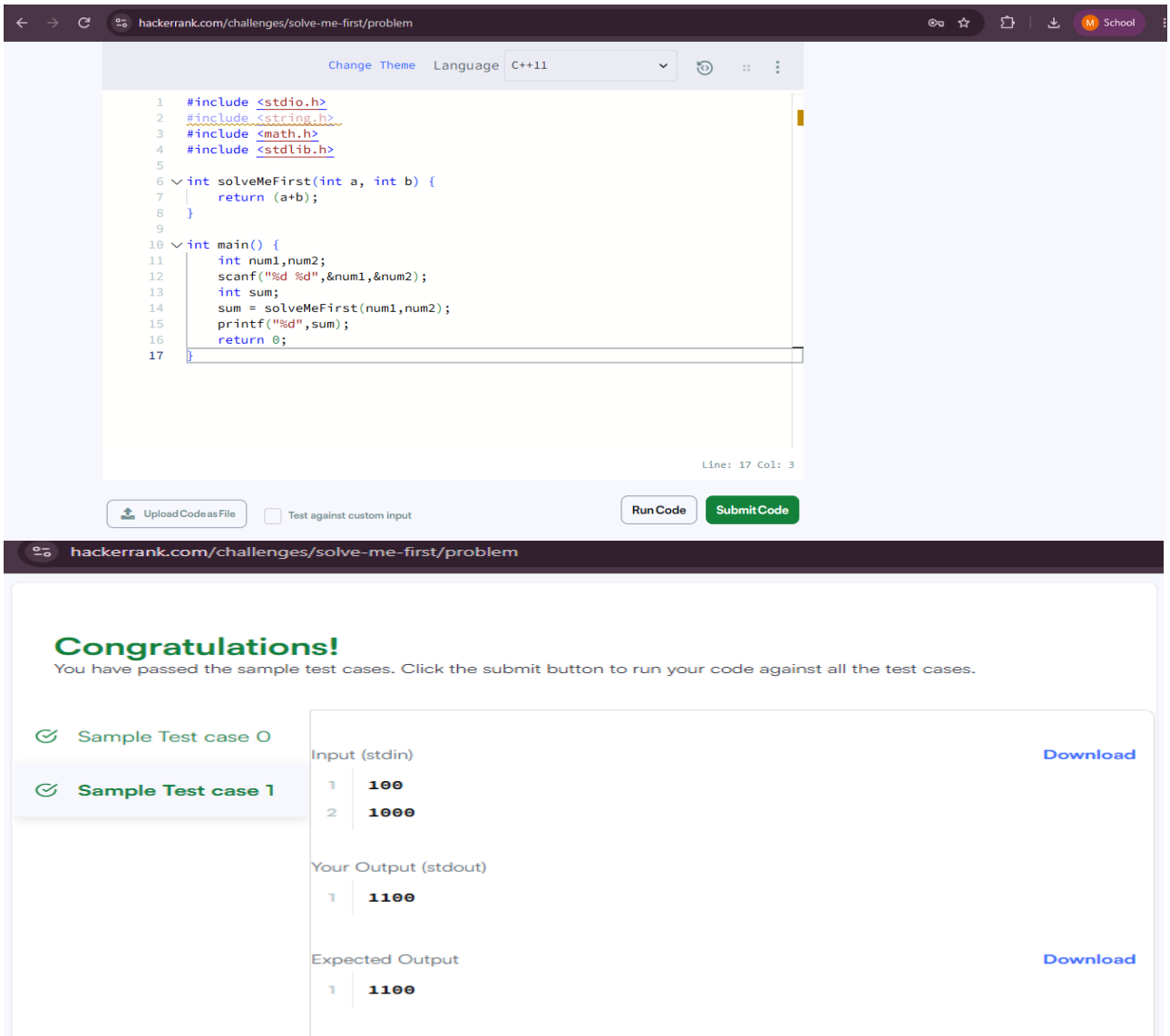
Answer:

```c
#include <stdio.h>
#include <string.h>
#include <math.h>
#include <stdlib.h>

int solveMeFirst(int a, int b) {
    return (a+b);

}

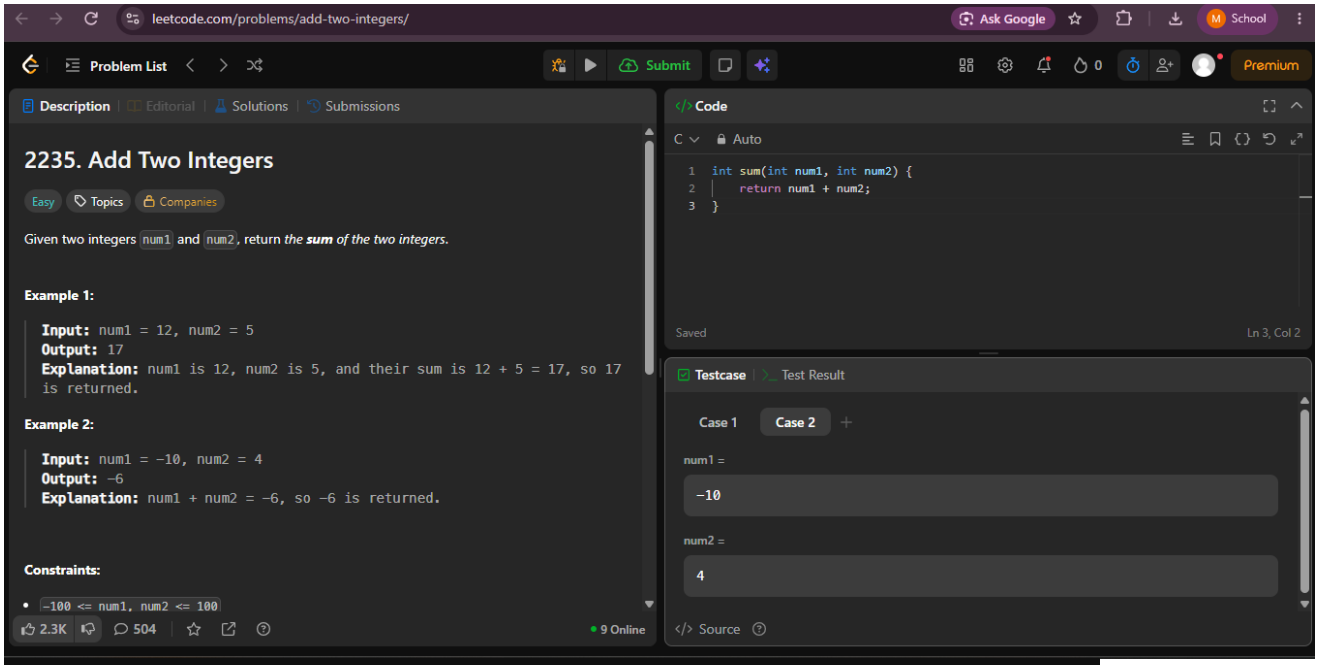
int main() {
    int num1,num2;
    scanf("%d %d",&num1,&num2);
    int sum;
    sum = solveMeFirst(num1,num2);
    printf("%d",sum);
    return 0;
}
```

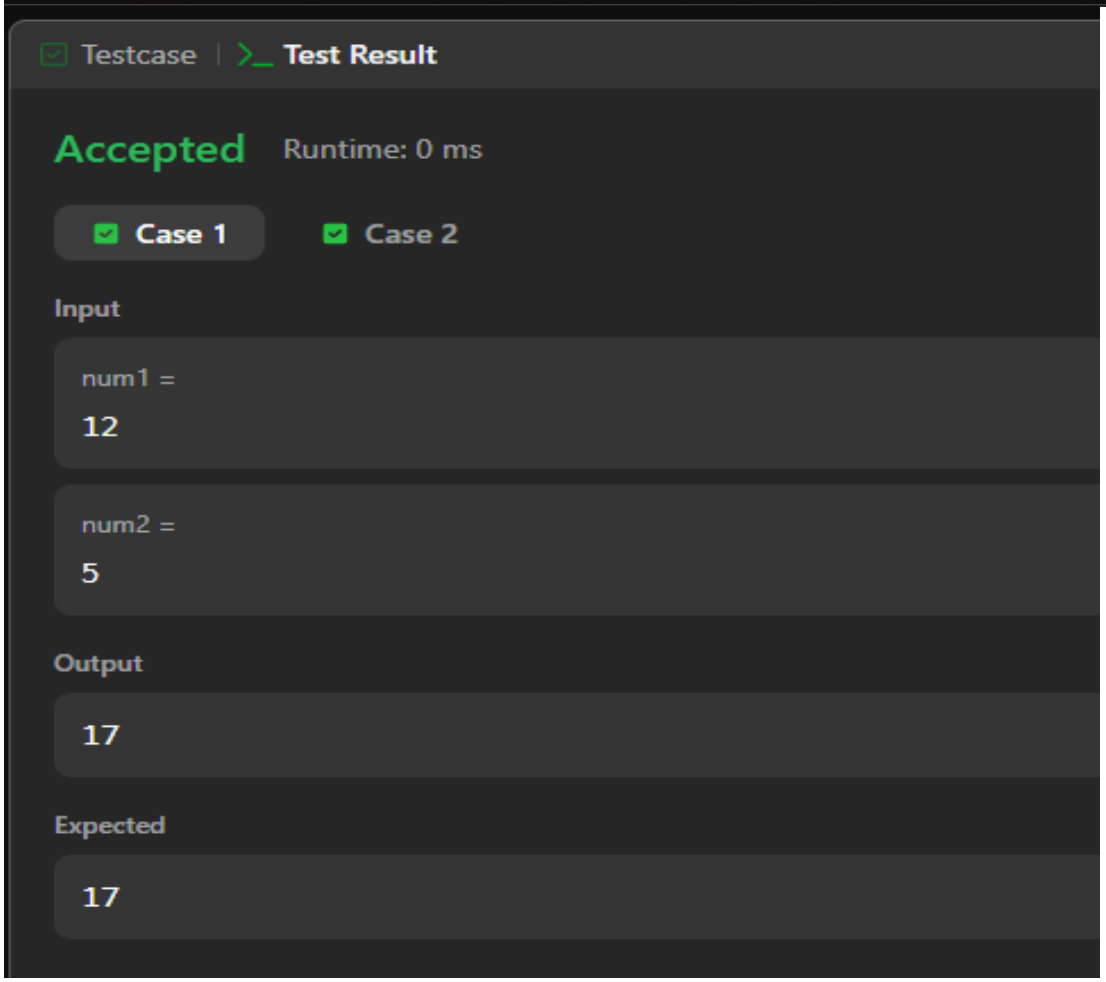


Question 3: 2235 — Add Two Integers

Answer:

```
int sum(int num1, int num2) {
    return(num1 + num2);
}
```

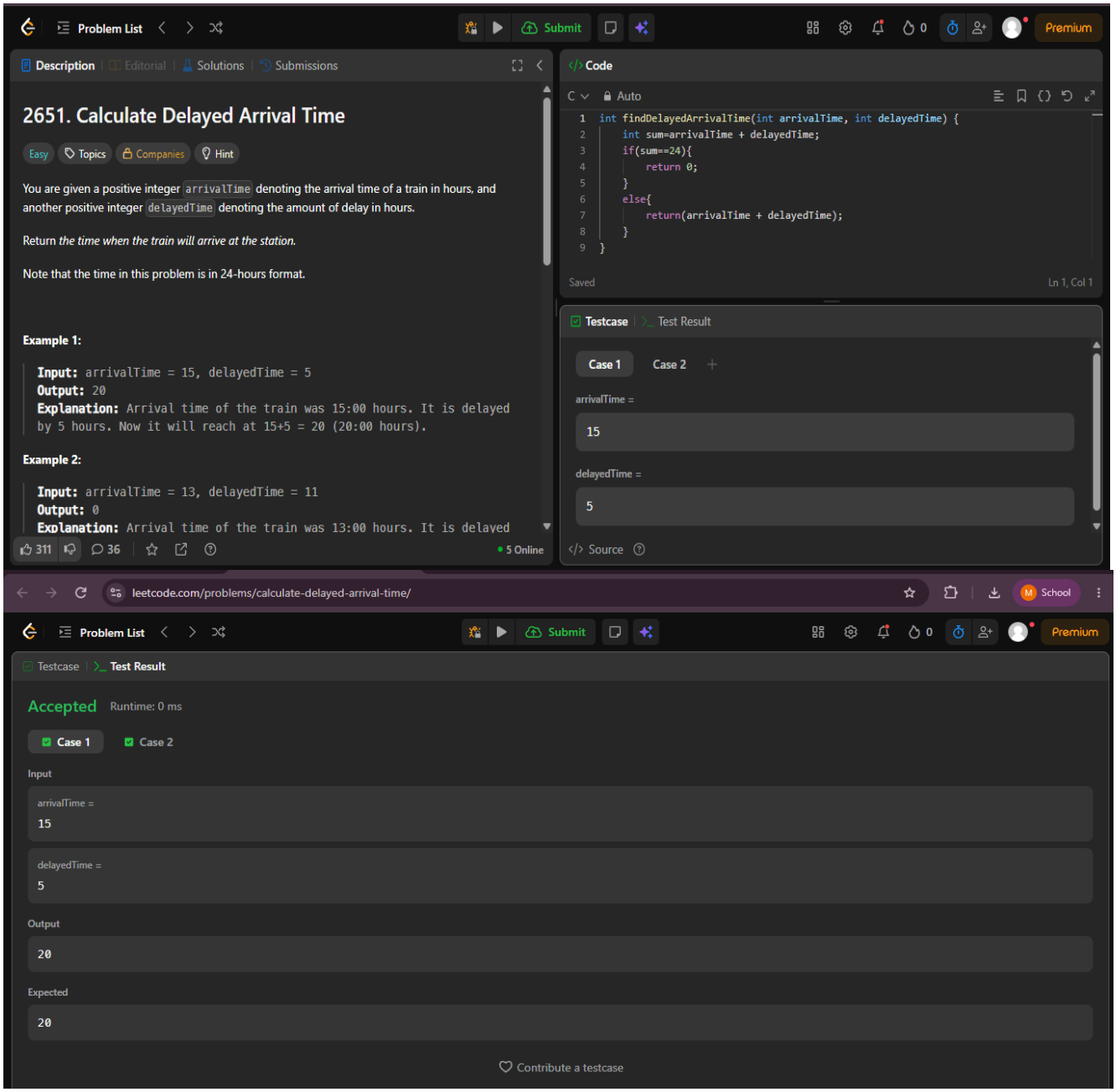




Question 4: 2651. Calculate Delayed Arrival Time

Answer:

```
int findDelayedArrivalTime(int arrivalTime, int delayedTime) {
    int sum=arrivalTime + delayedTime;
    if(sum==24){
        return 0;
    }
    else{
        return(arrivalTime + delayedTime);
    }
}
```



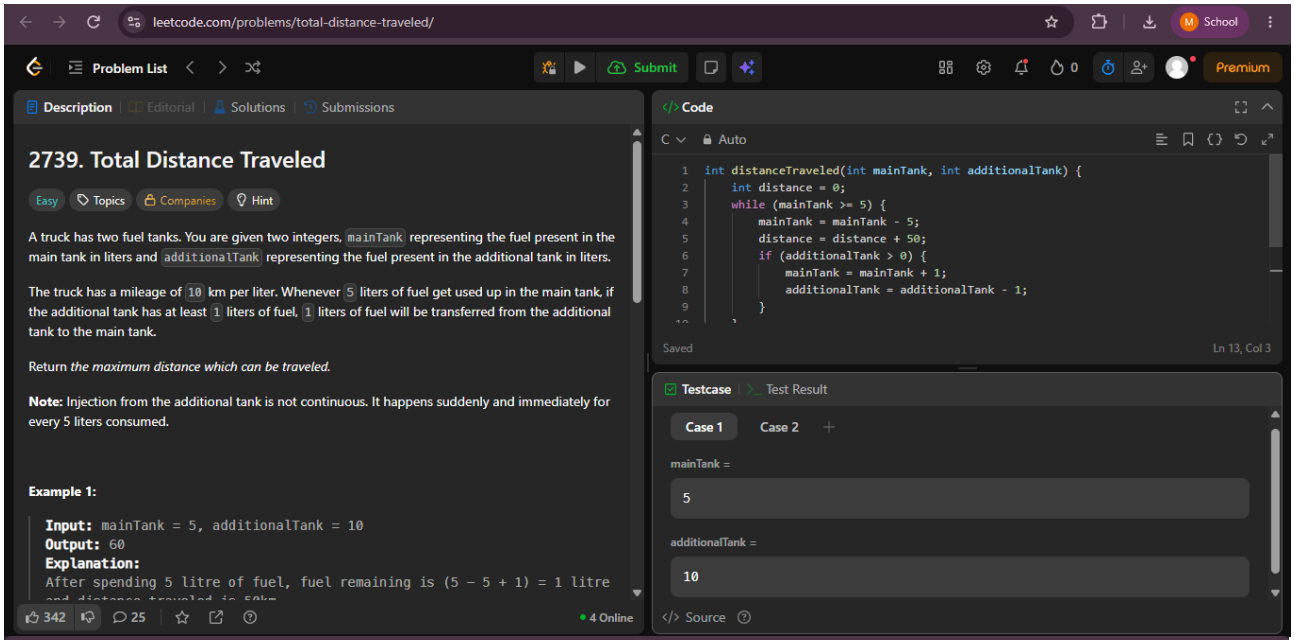
Question 5: 2739. Total Distance Traveled

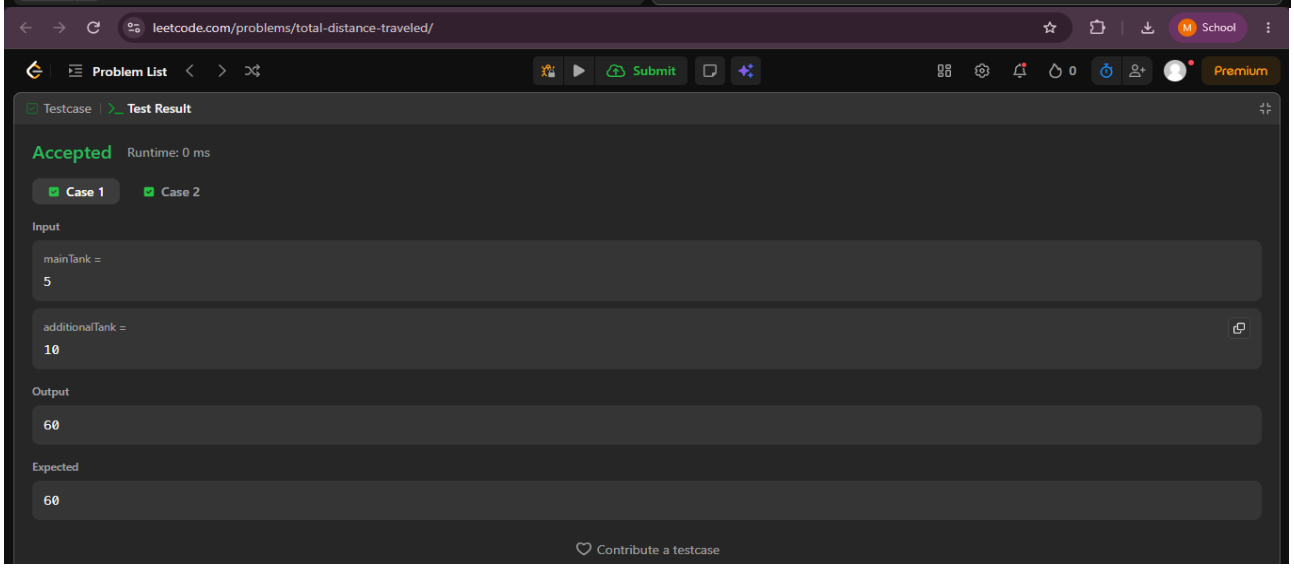
Answer:

```
int distanceTraveled(int mainTank, int additionalTank) {
    int distance = 0;
    while (mainTank >= 5) {
        mainTank = mainTank - 5;
        distance = distance + 50;

        if (additionalTank > 0) {
            mainTank = mainTank + 1;
            additionalTank = additionalTank - 1;
        }
    }

    distance = distance + mainTank * 10;
    return distance;
}
```



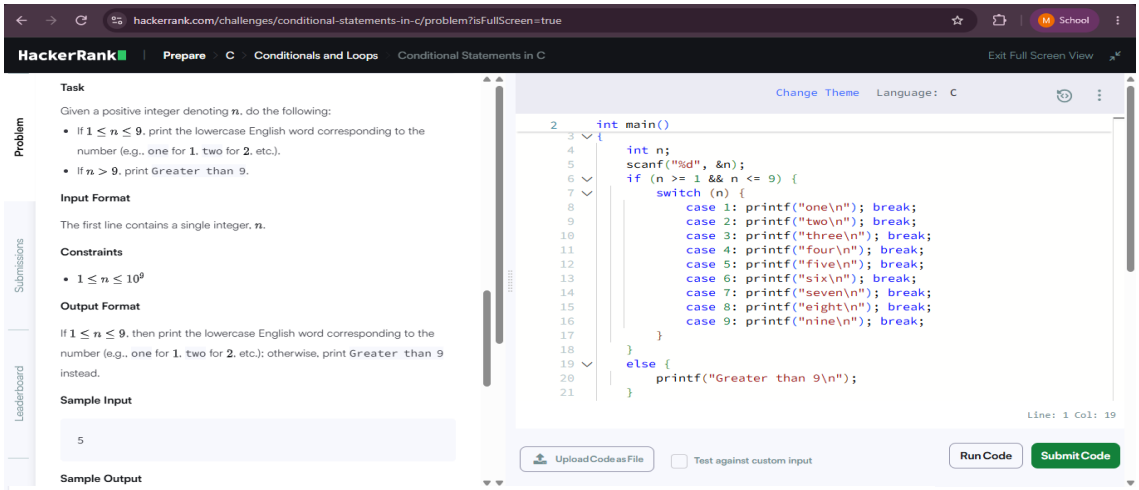


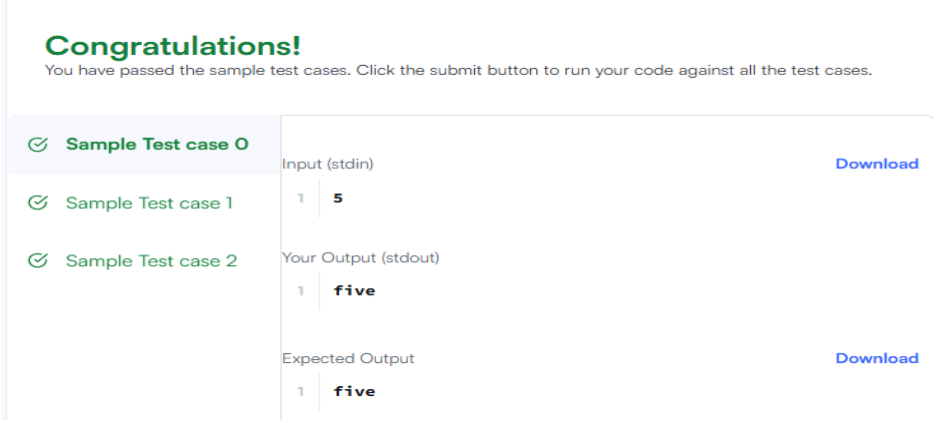
Module 2 — Conditional statements and operators

Question 6: Conditional Statements in C

Answer:

```c
#include <stdio.h>
int main()
{
    int n;
    scanf("%d", &n);
    if (n >= 1 && n <= 9) {
        switch (n) {
            case 1: printf("one\n"); break;
            case 2: printf("two\n"); break;
            case 3: printf("three\n"); break;
            case 4: printf("four\n"); break;
            case 5: printf("five\n"); break;
            case 6: printf("six\n"); break;
            case 7: printf("seven\n"); break;
            case 8: printf("eight\n"); break;
            case 9: printf("nine\n"); break;
        }
    }
    else {
        printf("Greater than 9\n");
    }
    return 0;
}
```





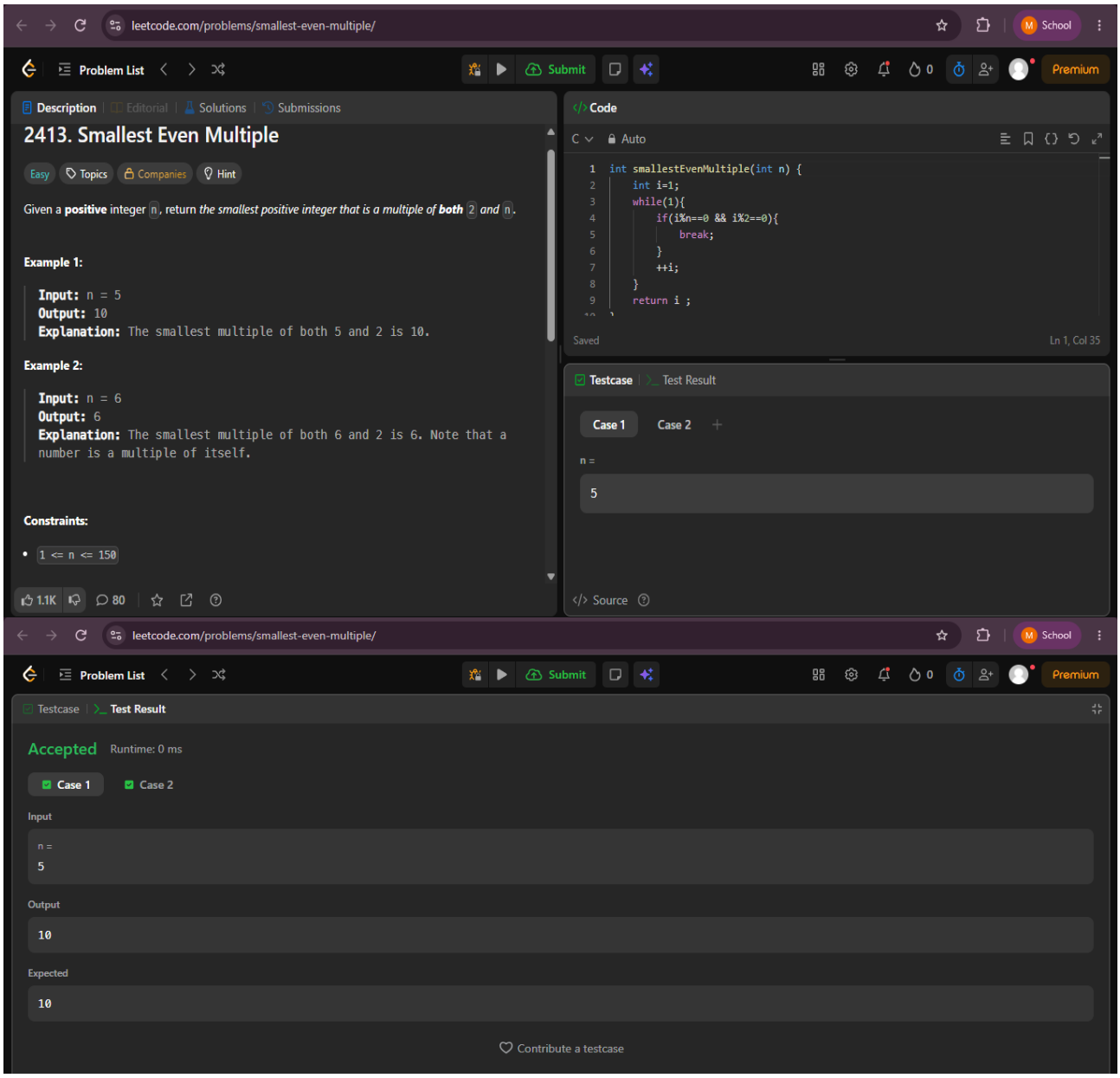
Question 7: 2413. Smallest Even Multiple

Answer:

```
int smallestEvenMultiple(int n) {
    int i=1;
    while(1){

        if(i%n==0 && i%2==0){
            break;
        }

        ++i;
    }
    return i ;
}
```



Question 8: Grading Students

Answer:

```
#include <stdio.h>
#include <stdlib.h>
```

```c
int* gradingStudents(int grades_count, int* grades, int* result_count) {
    *result_count = grades_count;
    int *result = malloc(grades_count * sizeof(int));
    for (int i = 0; i < grades_count; i++) {
        int grade = grades[i];
        int diff = 5 - (grade % 5);
        if (grade >= 38 && diff < 3) {
            grade = grade + diff;
        }
        result[i] = grade;
    }
    return result;
}

int main() {
    char *path = getenv("OUTPUT_PATH");
    FILE *fptr = path ? fopen(path, "w") : stdout;

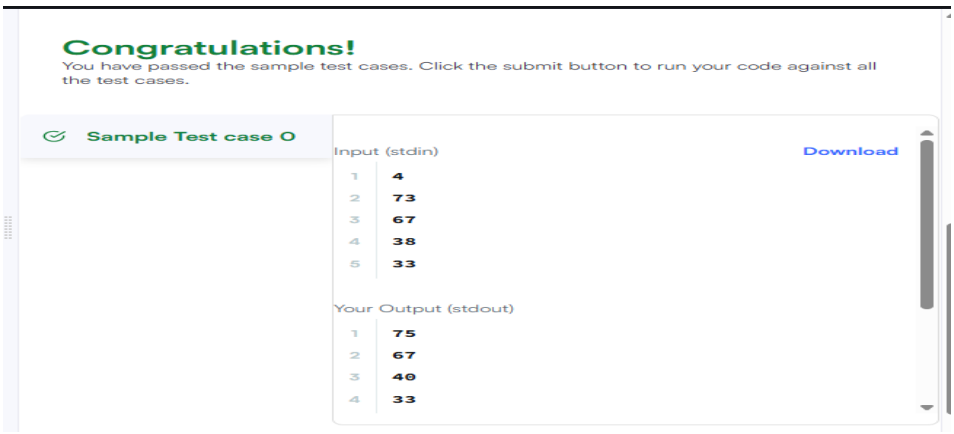
    int n;
    scanf("%d", &n);

    int *grades = malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        scanf("%d", &grades[i]);
    }

    int result_count;
    int *result = gradingStudents(n, grades, &result_count);

    for (int i = 0; i < result_count; i++) {
        fprintf(fptr, "%d\n", result[i]);
    }

    free(grades);
    free(result);
    if (path) fclose(fptr);
    return 0;
}
```



Question 9: Roy and Profile Picture

HackerEarth

Answer:

```c
#include <stdio.h>

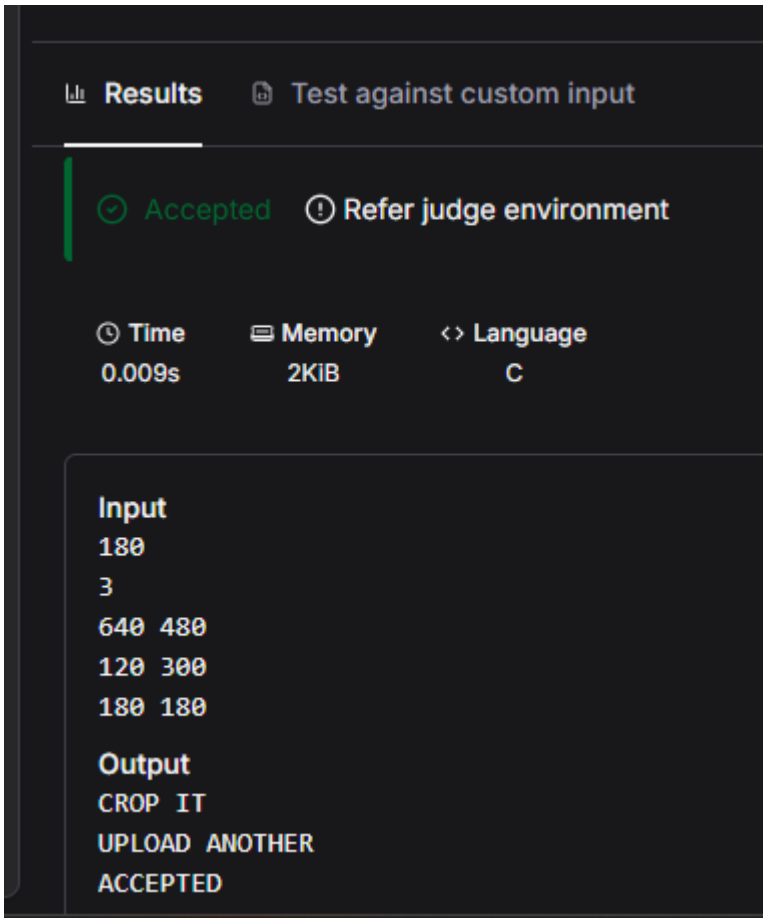
int main() {
    int L, N, W, H;

    scanf("%d", &L);
    scanf("%d", &N);

    for (int i = 0; i < N; i++) {
        scanf("%d %d", &W, &H);

        if (W < L || H < L) {
            printf("UPLOAD ANOTHER\n");
        }
        else if (W == H) {
            printf("ACCEPTED\n");
        }
        else {
            printf("CROP IT\n");
        }
    }

    return 0;
}
```

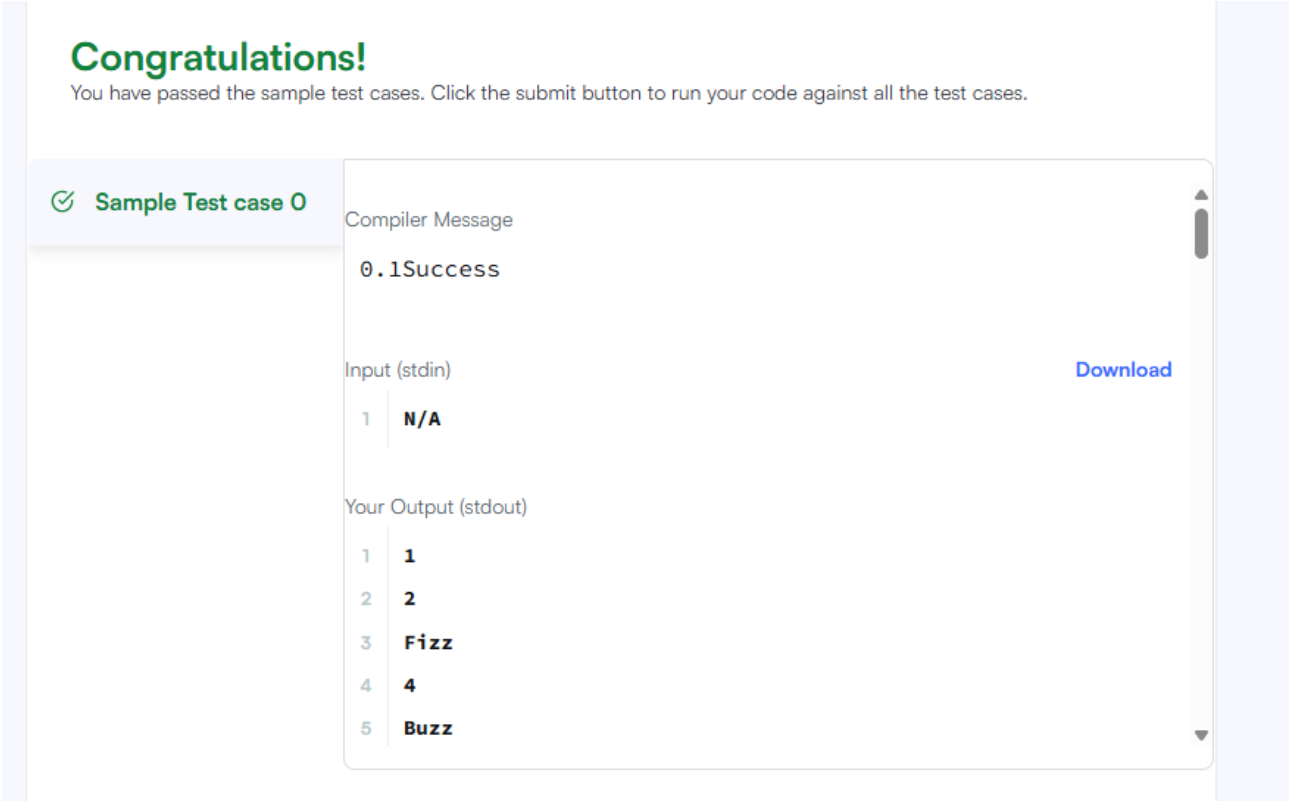


Question 10: FizzBuzz

HackerRank

Answer:

```
#include<stdio.h>
int main(){
    for(int i=1 ; i<=100 ; i++){
        if(i%3==0 && i%5==0){
            printf("FizzBuzz\n");
        }
        else if(i%3==0){
            printf("Fizz\n");
        }
        else if(i%5==0){
            printf("Buzz\n");
        }
        else{
            printf("%d\n",i);
        }
    }
    return 0;
}
```



Module 3 — Loops and repeated computation

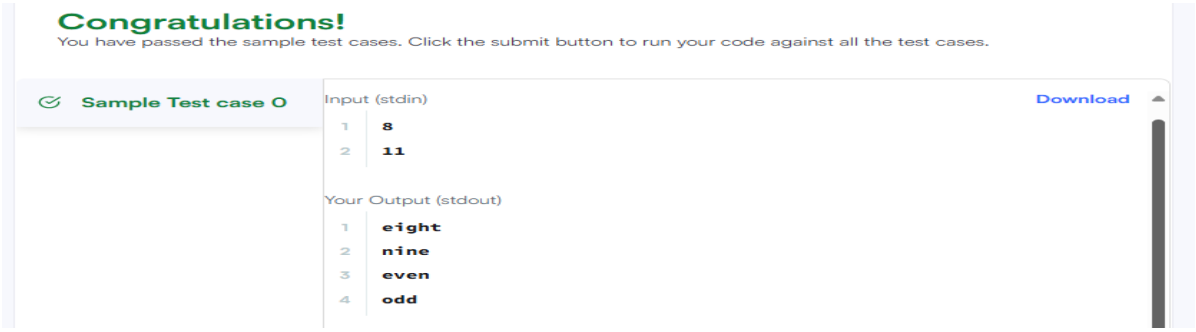
Question 11: For Loop in C

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Answer:

```
int main()
{
    int n,a, b;
    scanf("%d\n%d", &a, &b);
    for(n=a ; n<=b ; n++){
        if (n>=1 && n<=9) {
            switch(n){
                case 1 : printf("one\n");
                    break;
                case 2 : printf("two\n");
                    break;
                case 3 : printf("three\n");
                    break;
                case 4 : printf("four\n");
                    break;
                case 5 : printf("five\n");
                    break;
                case 6 : printf("six\n");
                    break;
                case 7 : printf("seven\n");
                    break;
                case 8 : printf("eight\n");
                    break;
                case 9 : printf("nine\n");
                    break;
                default:
                    printf("error");
            }
        }
        else if(n>9){
            if(n%2==0){
                printf("even\n");
            }
            else{
                printf("odd\n");
            }
        }
    }

    return 0;
}
```

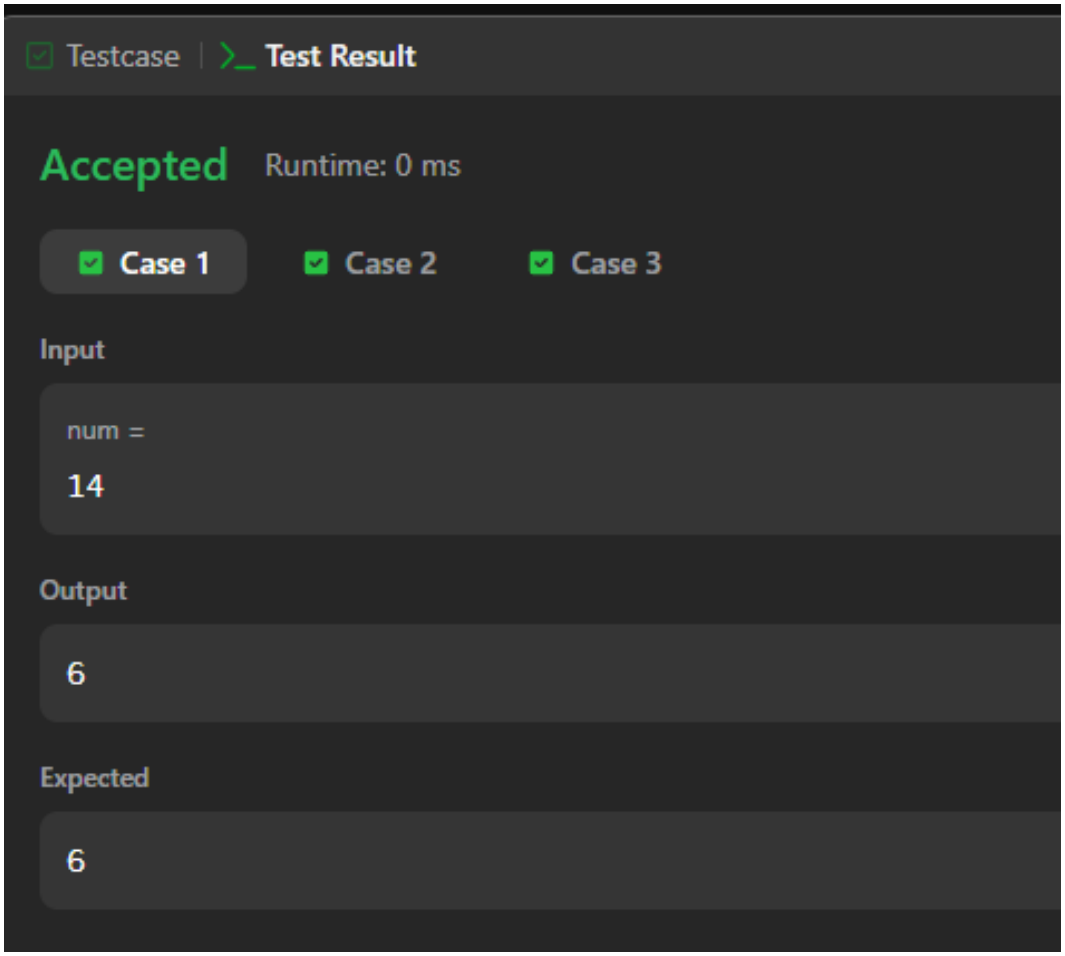


Question 12: 1342. Number of Steps to Reduce a Number to Zero

LeetCode

Answer:

```
int numberOfSteps(int num) {
    int count=0;
    while(num!=0){
        if(num%2==0){
            num=num/2;
        }
        else{
            num=num-1;
        }
        count ++;
    }
    return count;
}
```



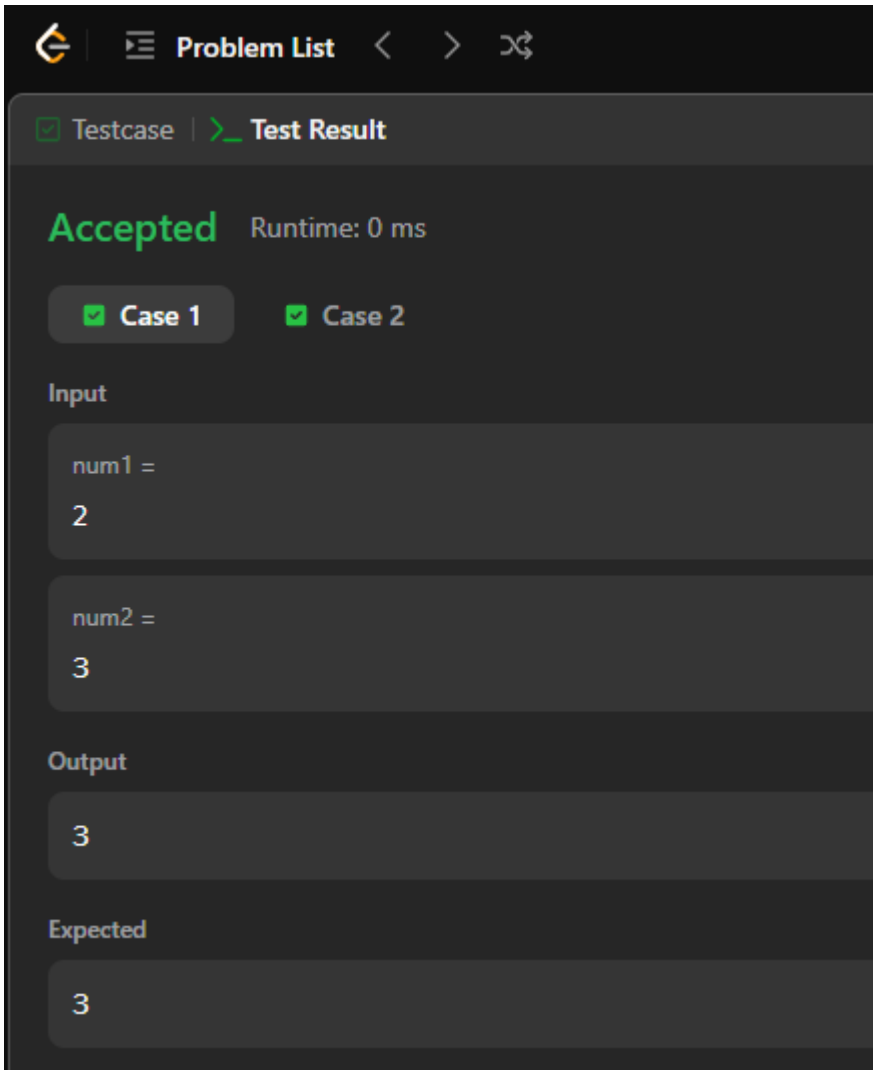
Question 13: 2169. Count Operations to Obtain Zero

LeetCode

Answer:

```
int countOperations(int num1, int num2) {
    int count=0,max;
    if(num1==0 || num2==0){
        return 0;
    }
    while(num1>0 && num2>0){
        max = (num1>num2)? num1 : num2;
        if (max == num1){
```

```
            num1=num1-num2;
        }
        else{
            num2=num2-num1;
        }
        count ++;
    }
    return count;
}
```

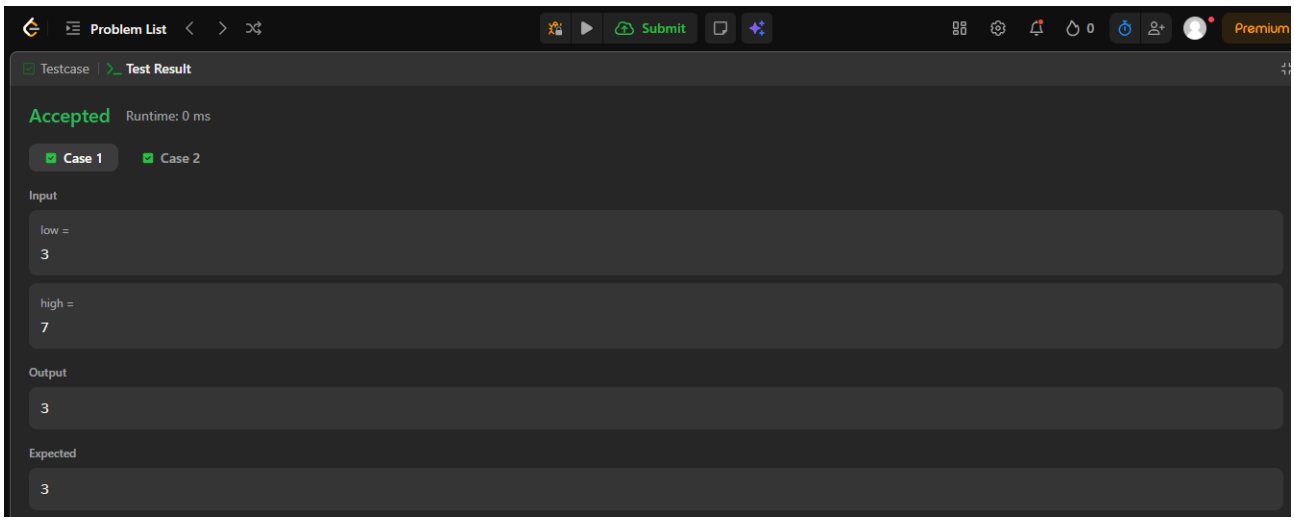


Question 14: 1523. Count Odd Numbers in an Interval Range

LeetCode

Answer:

```
int countOdds(int low, int high){
    return (high + 1) / 2 - low / 2;
}
```

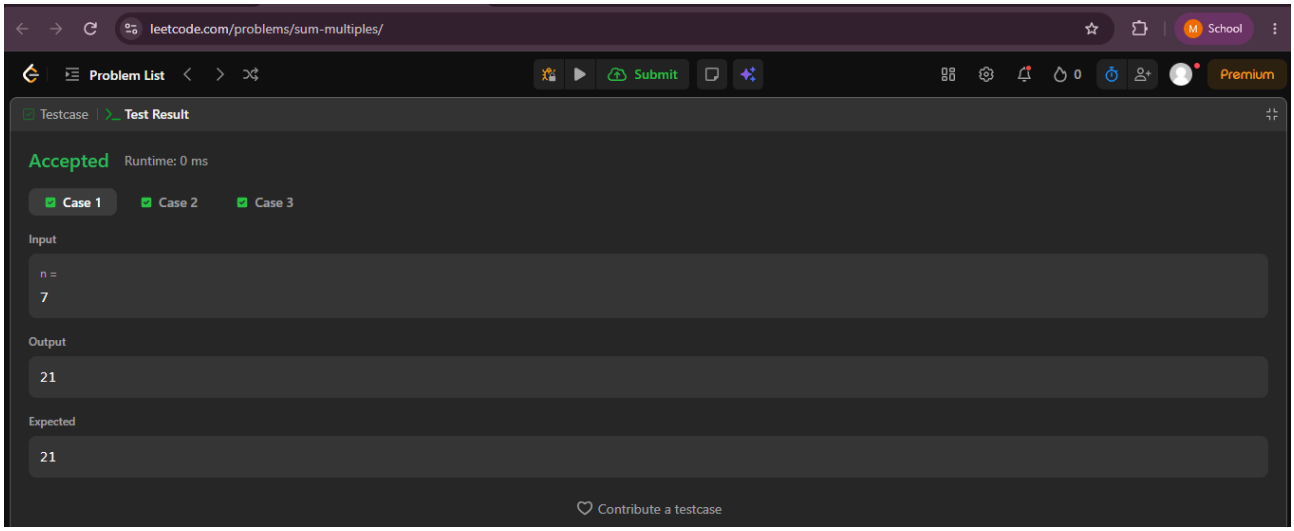


Question 15: 2652. Sum Multiples

LeetCode

Answer:

```
int sumOfMultiples(int n) {
    int sum=0;
    for(int i=1;i<=n;i++){
        if(i%3==0|| i%5==0 || i%7==0){
            sum=sum+i;
        }
    }
    return sum;
}
```



Question 16: Number Line Jumps

HackerRank

Answer:

```
char* kangaroo(int x1, int v1, int x2, int v2) {

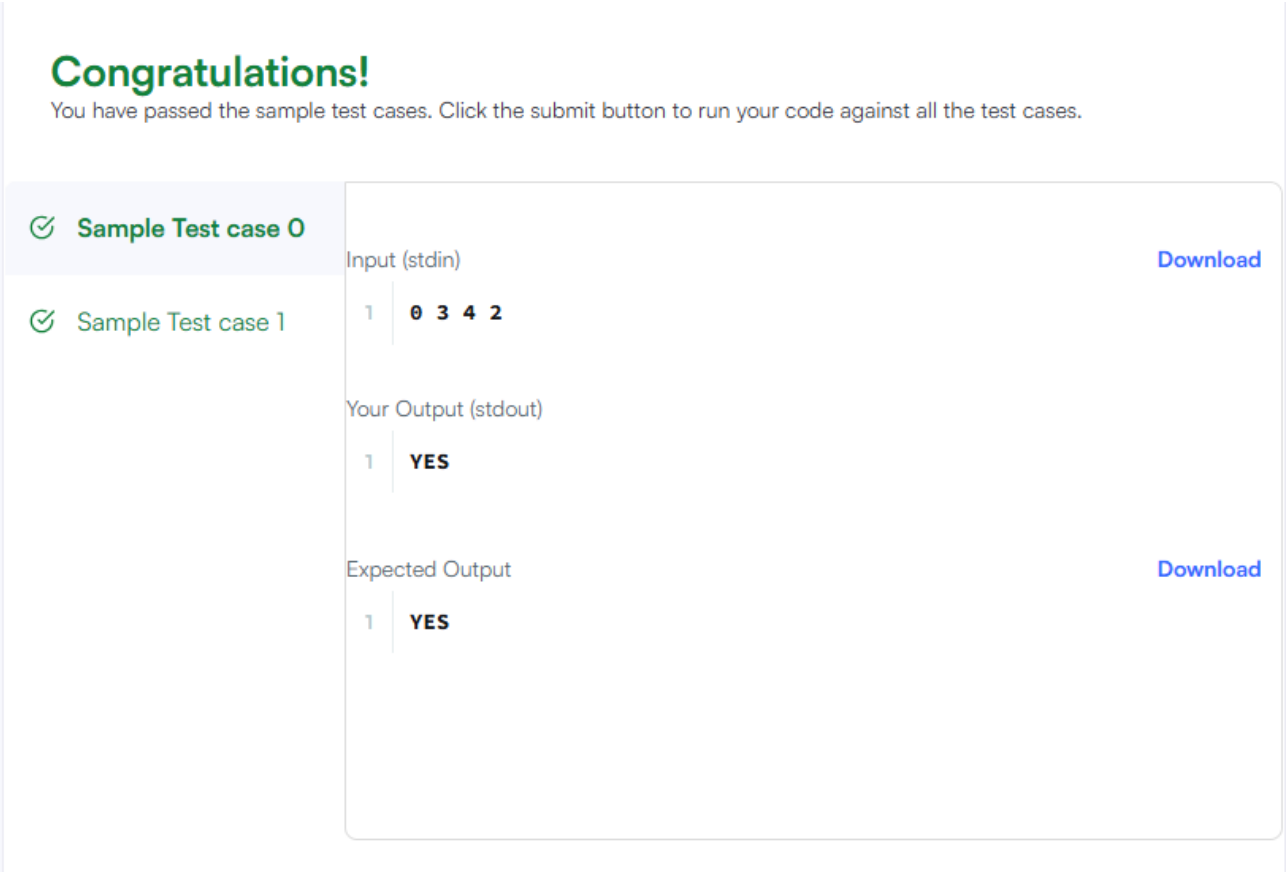
    char *yes = "YES";
    char *no = "NO";

    if(v1 <= v2) return no;
```

```
    int n = (x2 - x1) / (v1 - v2);
    int m = (x2 - x1) % (v1 - v2);

    if((n>0) && (m==0))
        return yes;
    else
        return no;

}
```



Module 4 — Digit extraction and number manipulation

Question 17: Digit Sum

HackerEarth

Answer:

```
#include <stdio.h>

int main() {
    int T, N, first, last, sum;

    scanf("%d", &T);

    while (T--) {
        scanf("%d", &N);

        last = N % 10;

        while (N >= 10) {
```

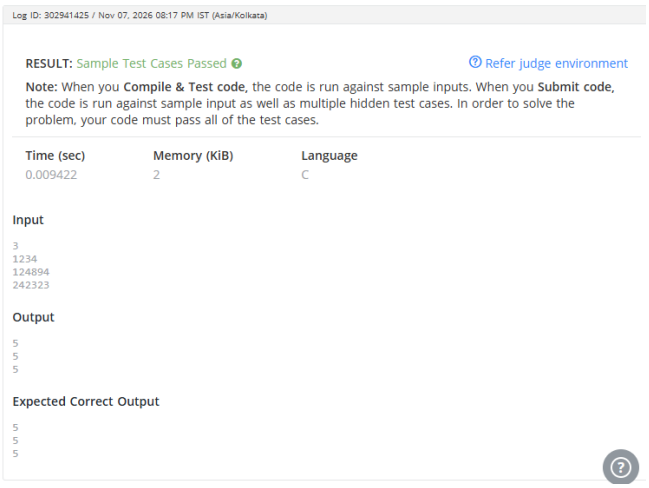
```
            N = N / 10;
        }

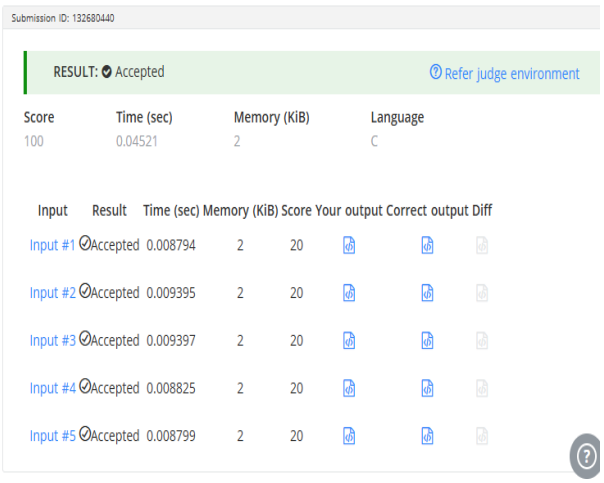
        first = N;

        sum = first + last;

        printf("%d\n", sum);
    }

    return 0;
}
```





Question 18: Find Digits

HackerRank

Answer:

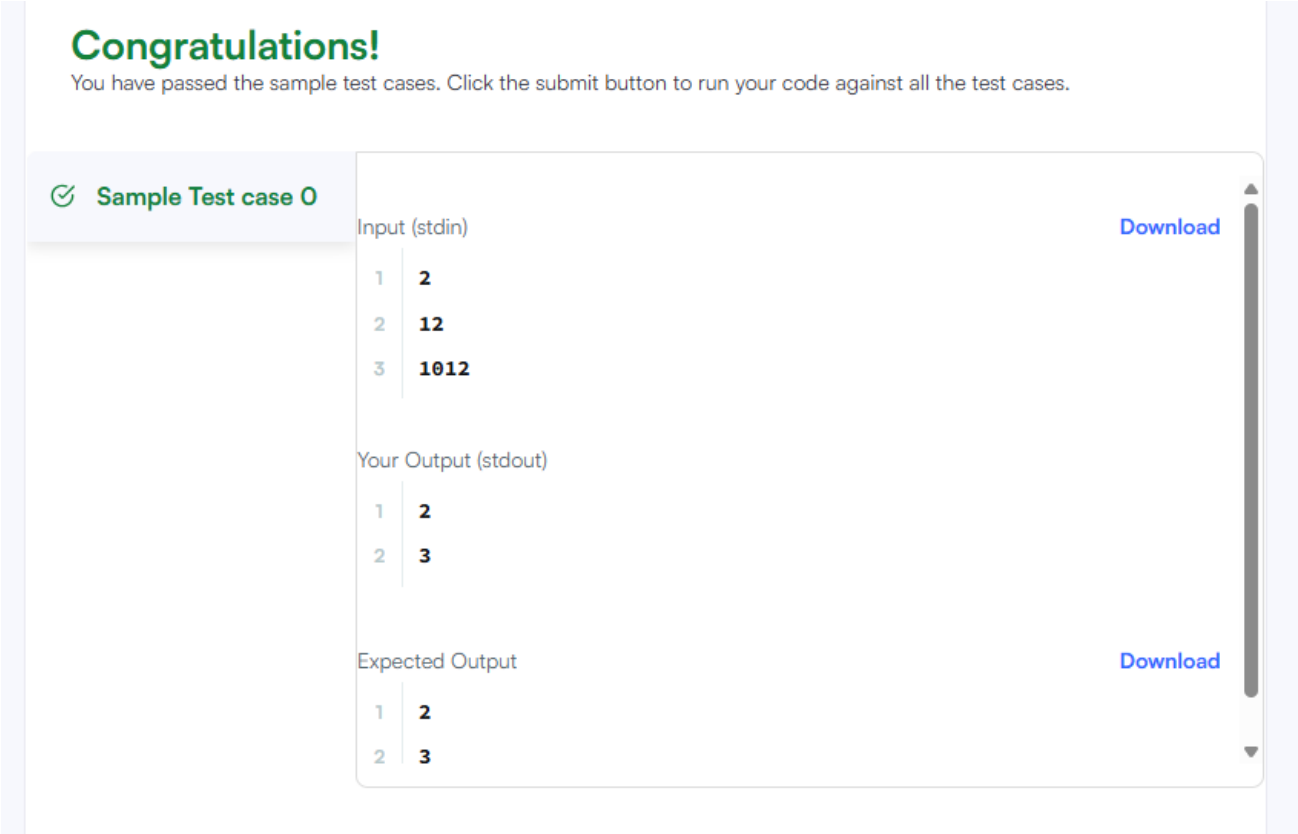
```
int findDigits(int n) {
    int sum = 0;
    int digit;
    int original = n;

    while (n > 0) {
        digit = n % 10;

        if (digit != 0 && original % digit == 0) {
            sum++;
        }

        n = n / 10;
    }

    return sum;
}
```

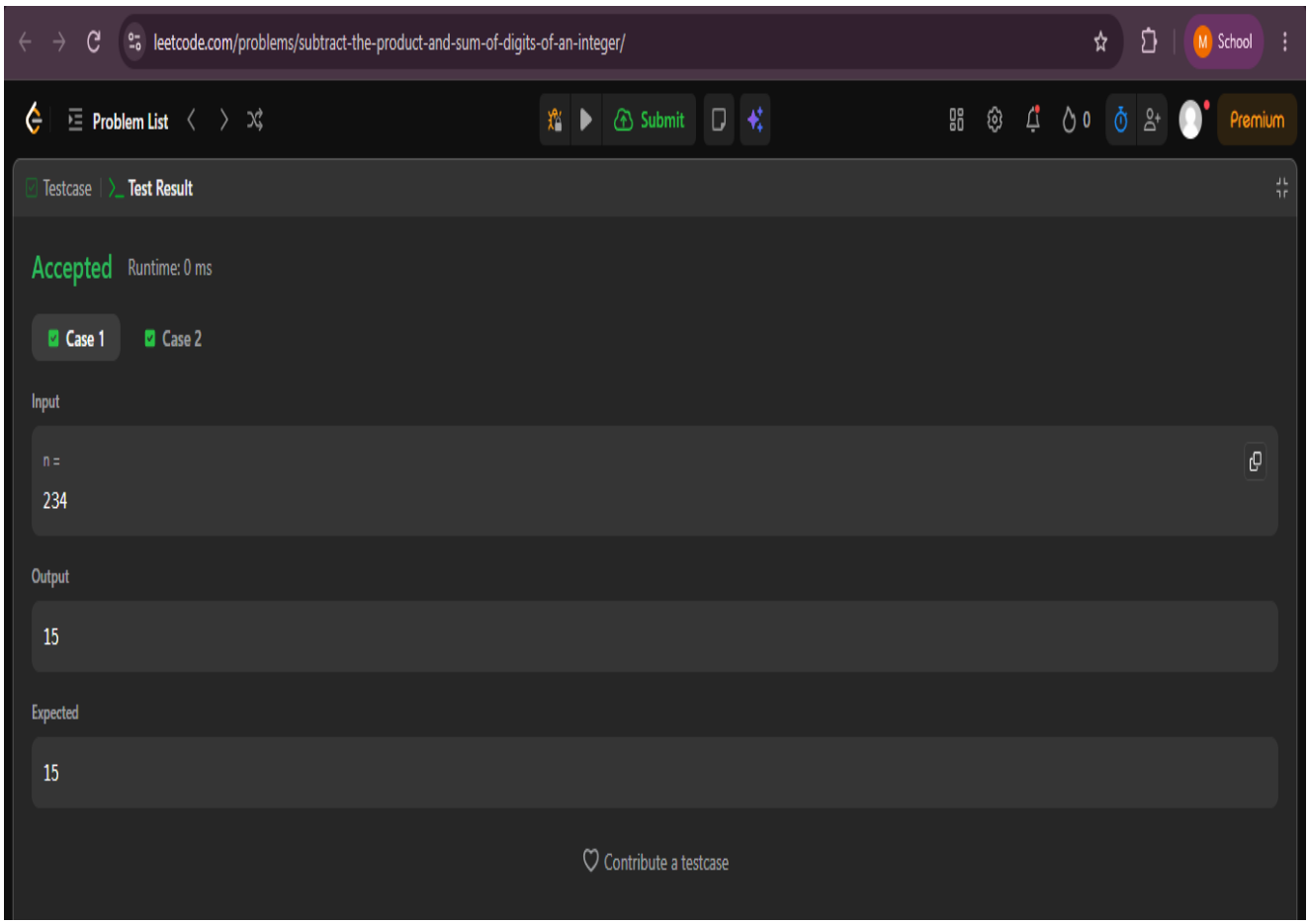


Question 19: 1281. Subtract the Product and Sum of Digits of an Integer

LeetCode

Answer:

```
int subtractProductAndSum(int n) {
    int prod=1,sum=0,org,digit,x;
    org=n;
    while(n>0){
        digit=n%10;
        sum=sum+digit;
        prod=prod*digit;
        n=n/10;
    }
    x=prod-sum;
    return x;
}
```



Question 20: 2520. Count the Digits That Divide a Number

LeetCode

Answer:

```
int countDigits(int num) {
    int original = num;
    int digit;
    int count = 0;

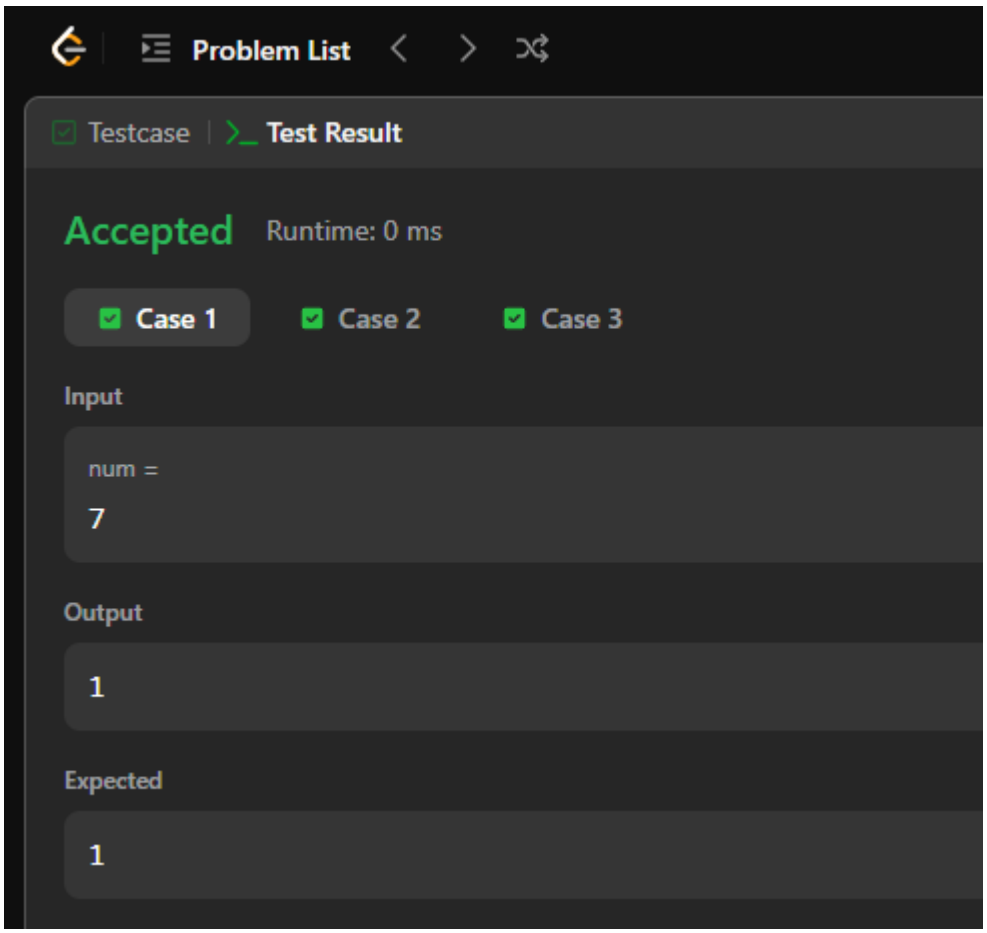
    while (num > 0) {
        digit = num % 10;

        if (digit != 0 && original % digit == 0) {
            count++;
        }

        num = num / 10;
    }

    return count;

}
```



Question 21: 9. Palindrome Number

LeetCode

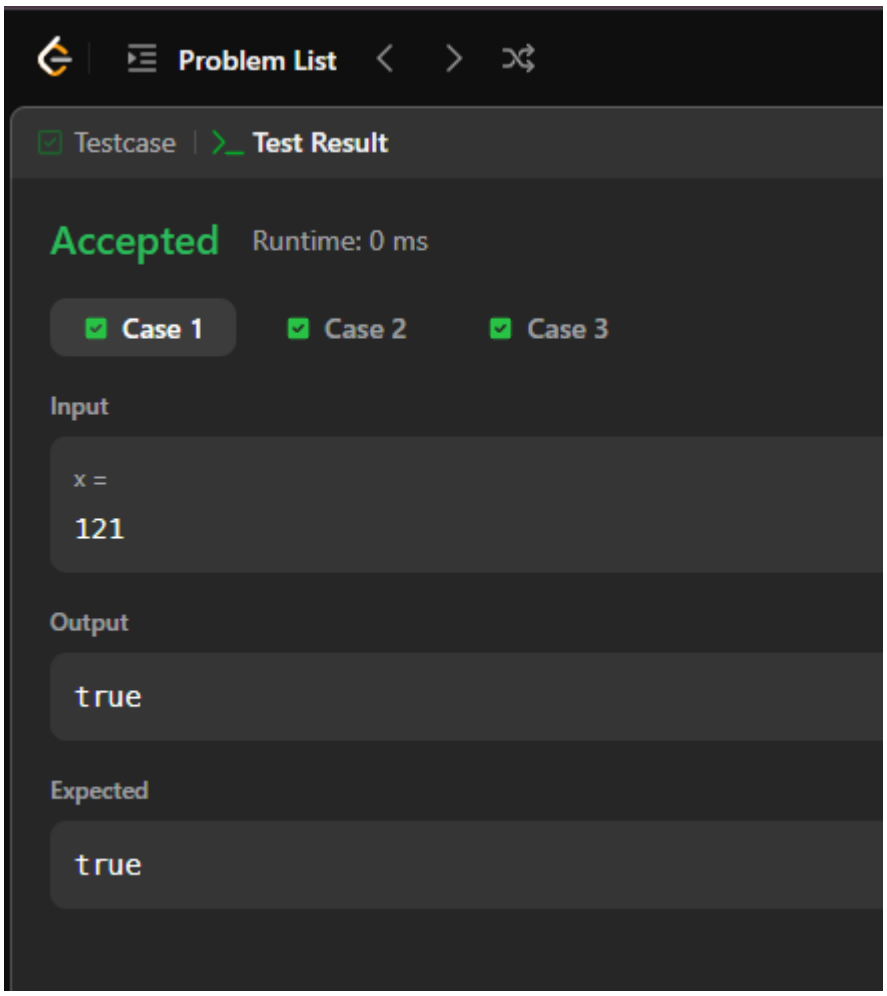
Answer:

```
bool isPalindrome(int x) {
    if (x < 0) {
        return false;
    }

    int original = x;
    long long reverse = 0;

    while (x > 0) {
        int digit = x % 10;
        reverse = reverse * 10 + digit;
        x = x / 10;
    }

    return original == reverse;
}
```



Question 22: 7. Reverse Integer

LeetCode

Answer:

```
#include <limits.h>

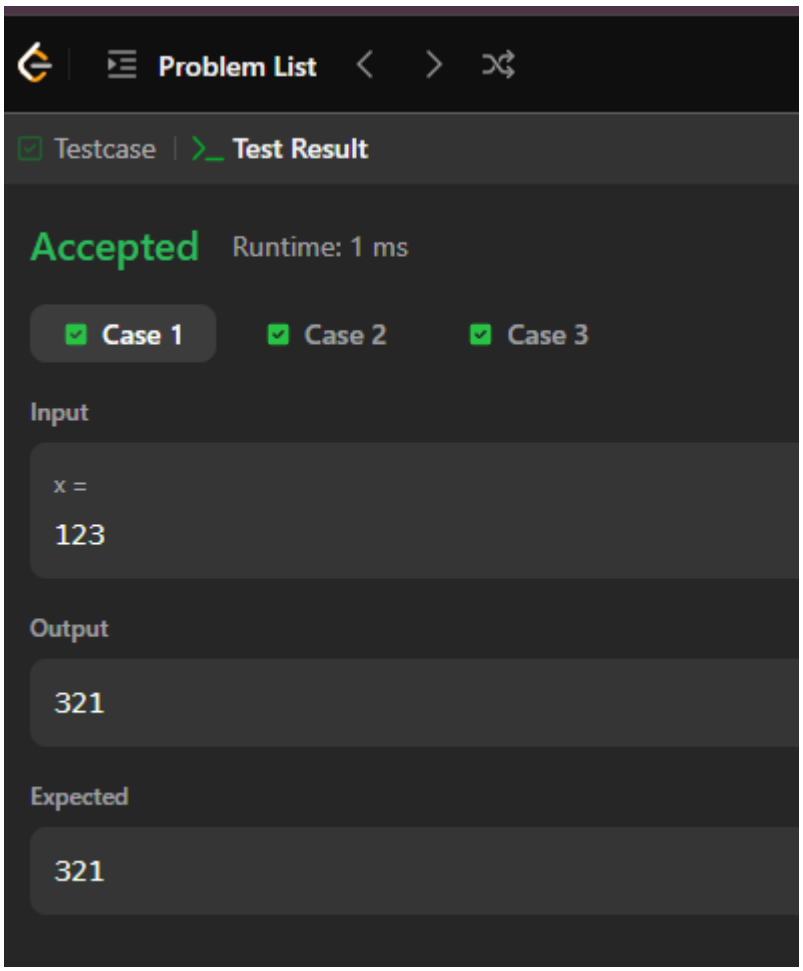
int reverse(int x) {
    int rev = 0;

    while (x != 0) {
        int digit = x % 10;
        x /= 10;

        if (rev > INT_MAX / 10 || (rev == INT_MAX / 10 && digit > 7)) {
            return 0;
        }
        if (rev < INT_MIN / 10 || (rev == INT_MIN / 10 && digit < -8)) {
            return 0;
        }

        rev = rev * 10 + digit;
    }

    return rev;
}
```



Module 5 — Factors, divisibility and mathematical loops

Question 23: 231. Power of Two

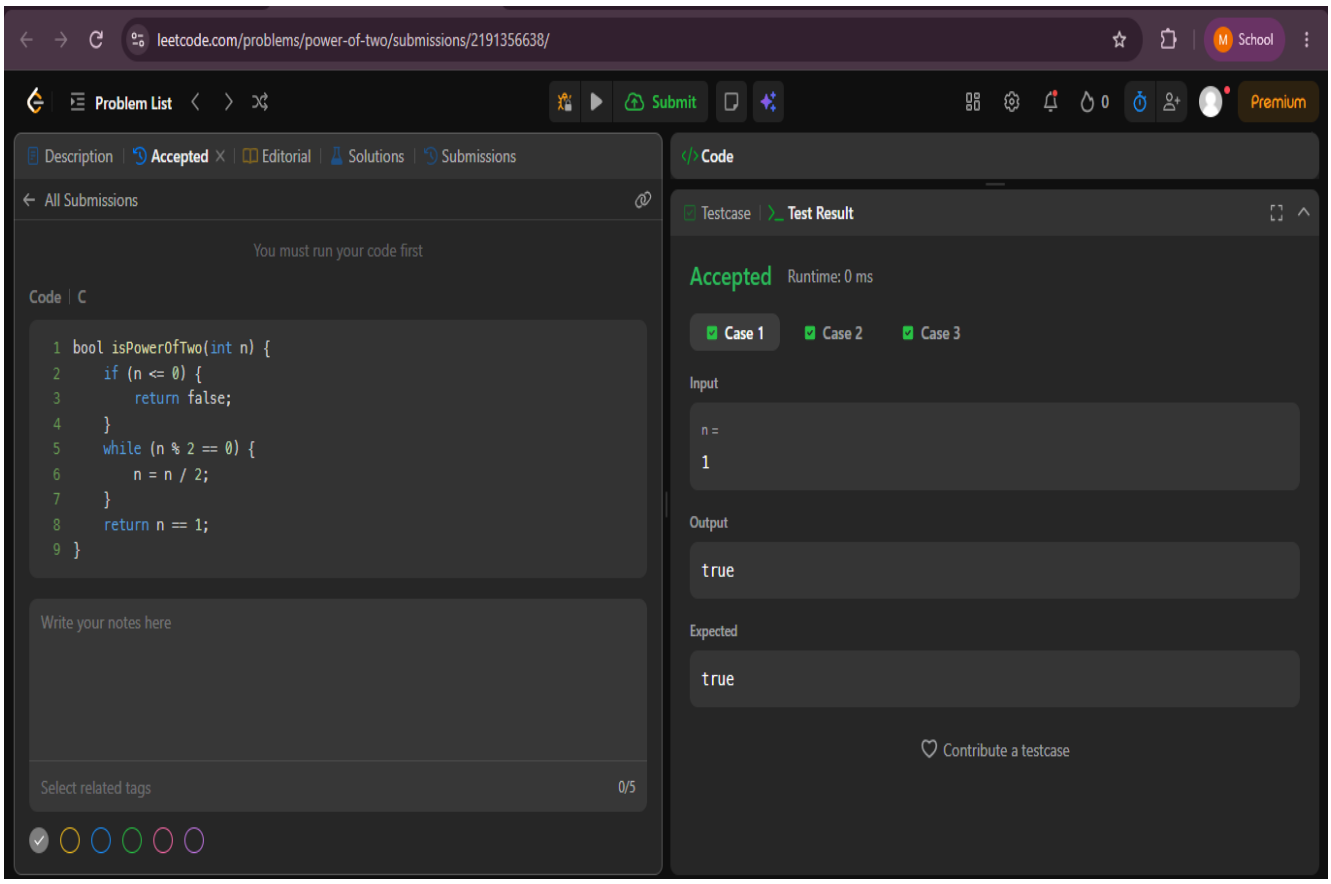
LeetCode

Answer:

```
bool isPowerOfTwo(int n) {
    if (n <= 0) {
        return false;
    }

    while (n % 2 == 0) {
        n = n / 2;
    }

    return n == 1;
}
```

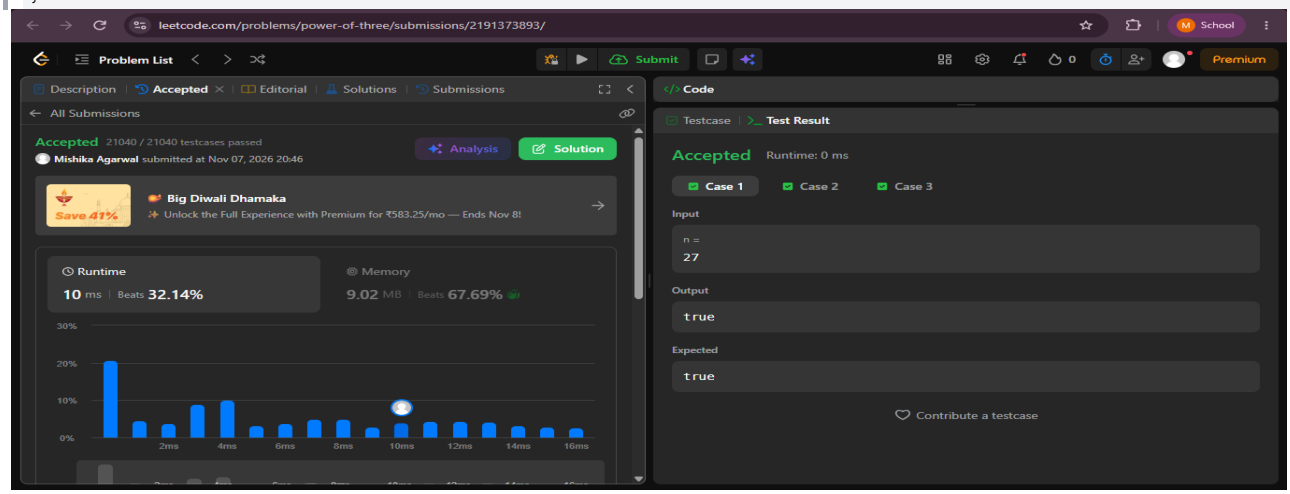


Question 24: 326. Power of Three

LeetCode

Answer:

```
bool isPowerOfThree(int n) {
    if (n <= 0) return false;
    while (n % 3 == 0) {
        n /= 3;
    }
    return n == 1;
}
```



Question 25: Factorial

HackerEarth

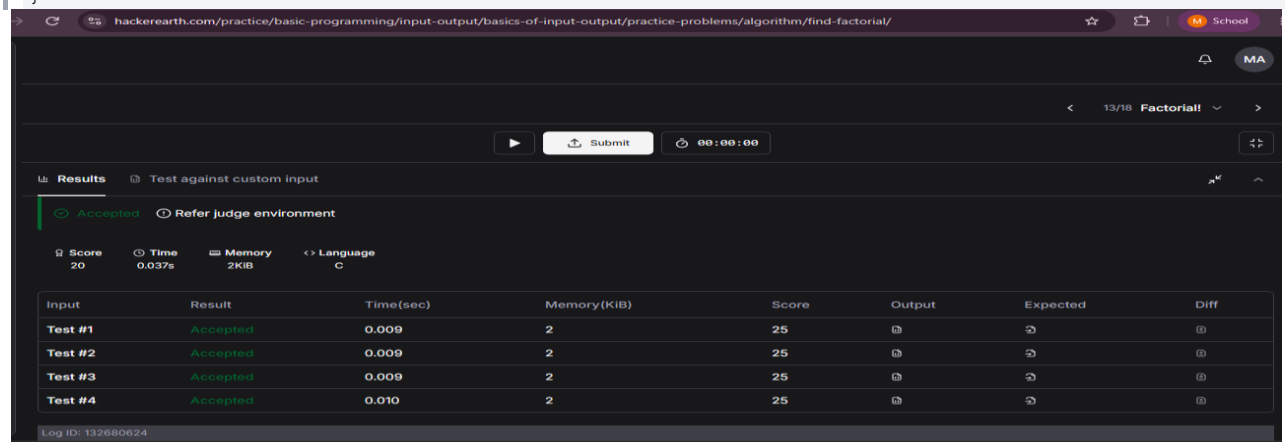
Answer:

```c
#include <stdio.h>

int main() {
    int num;
    scanf("%d", &num);

    int fact = 1;
    for (int i = 2; i <= num; i++) {
        fact *= i;
    }

    printf("%d\n", fact);
    return 0;
}
```



Question 26: Count Divisors

HackerEarth

Answer:

```c
#include <stdio.h>

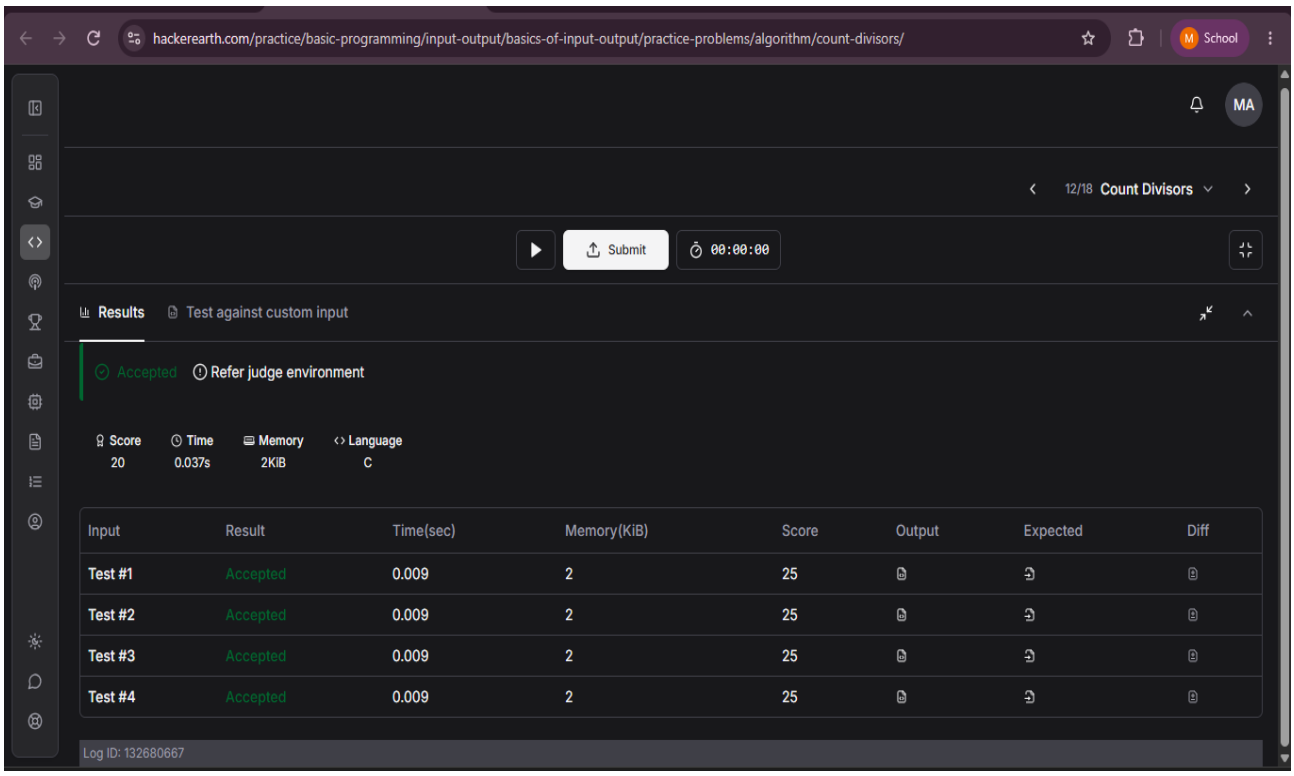
int main() {
    int l, r, k;
    int count = 0;

    scanf("%d %d %d", &l, &r, &k);

    for (int i = l; i <= r; i++) {
        if (i % k == 0) {
            count++;
        }
    }

    printf("%d\n", count);

    return 0;
}
```



Question 27: Prime Numbers

HackerEarth

Answer:

```
#include <stdio.h>

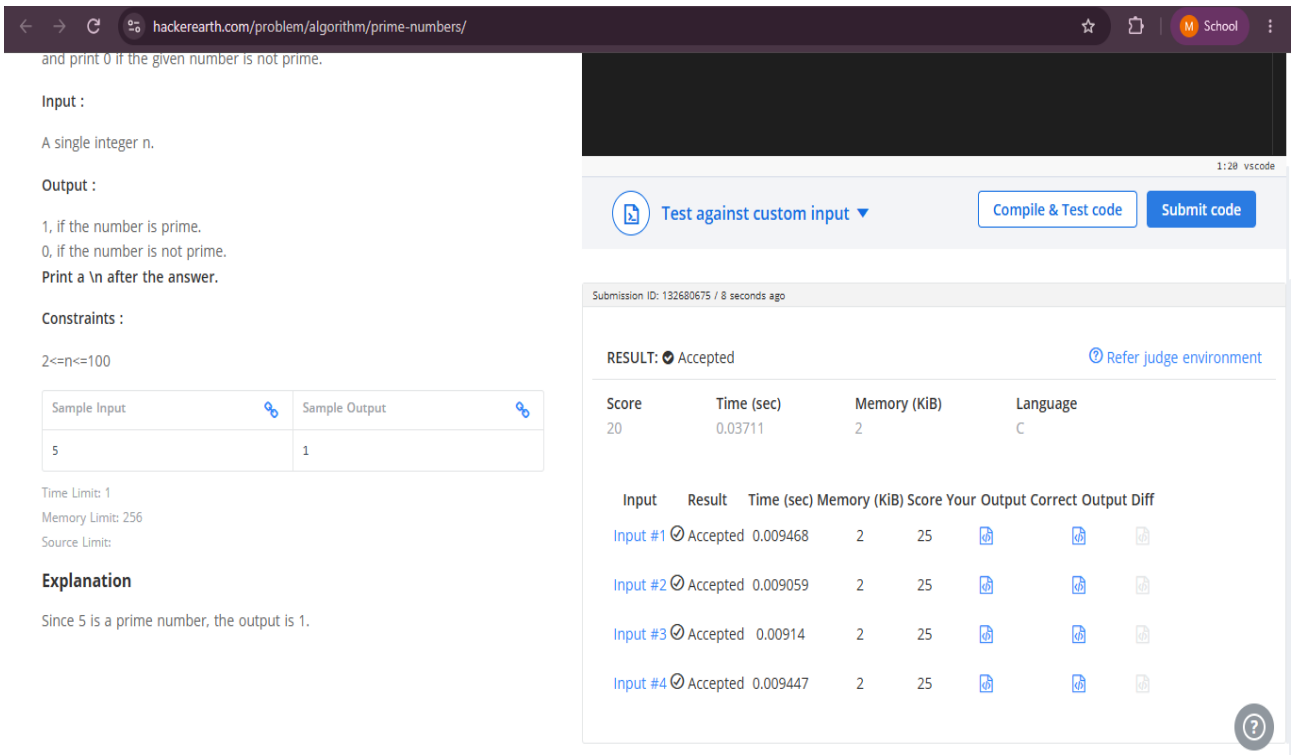
int main() {
    int n, flag = 1;

    scanf("%d", &n);

    for (int i = 2; i < n; i++) {
        if (n % i == 0) {
            flag = 0;
            break;
        }
    }

    printf("%d\n", flag);

    return 0;
}
```



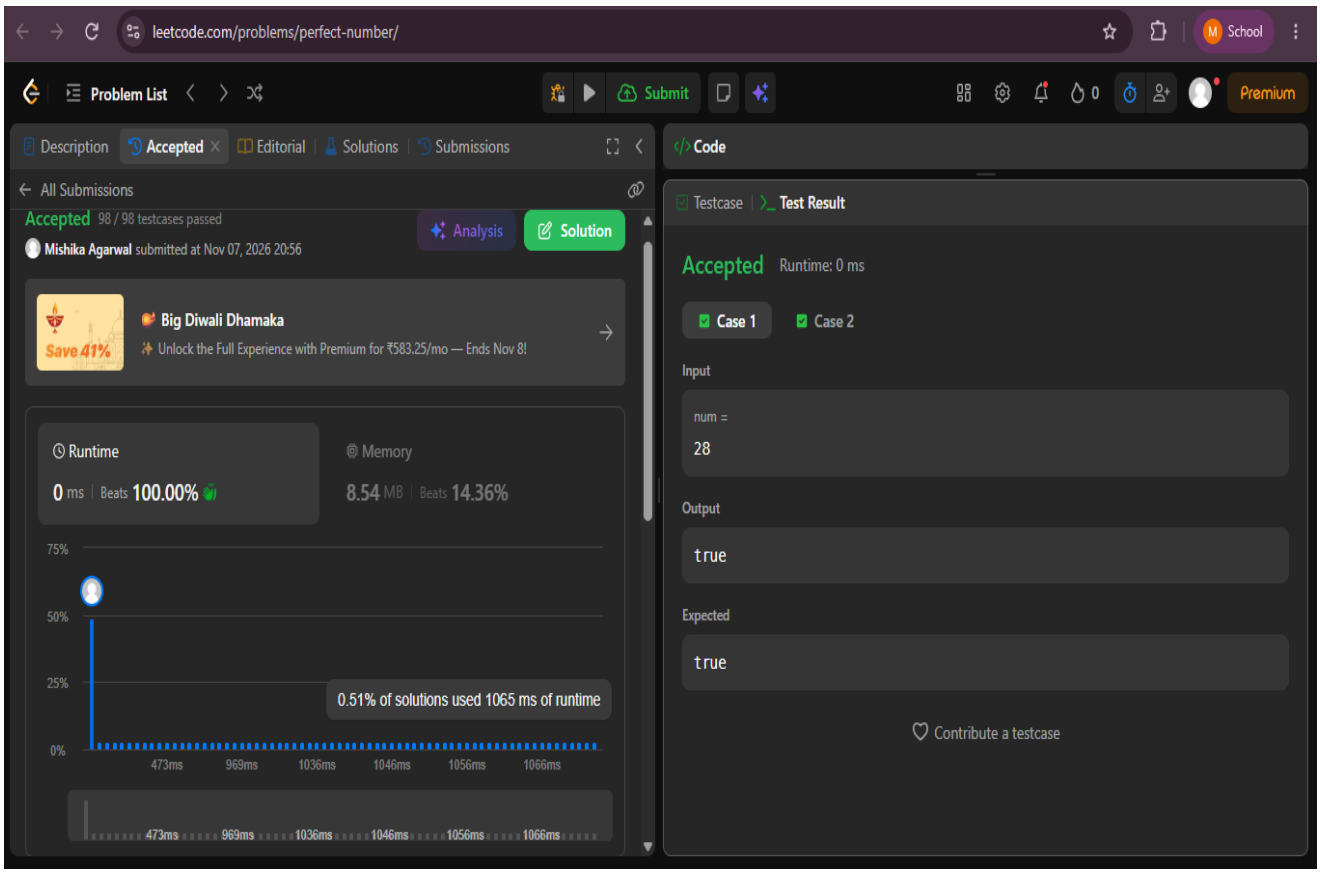
Question 28: Perfect Numbers

LeetCode

Answer:

```
bool checkPerfectNumber(int num) {
    if (num <= 1) return false;

    int sum = 1;  // 1 is a divisor of every number > 1
    for (int i = 2; (long long)i * i <= num; i++) {
        if (num % i == 0) {
            sum += i;
            if (i != num / i) {
                sum += num / i;
            }
        }
    }
    return sum == num;
}
```



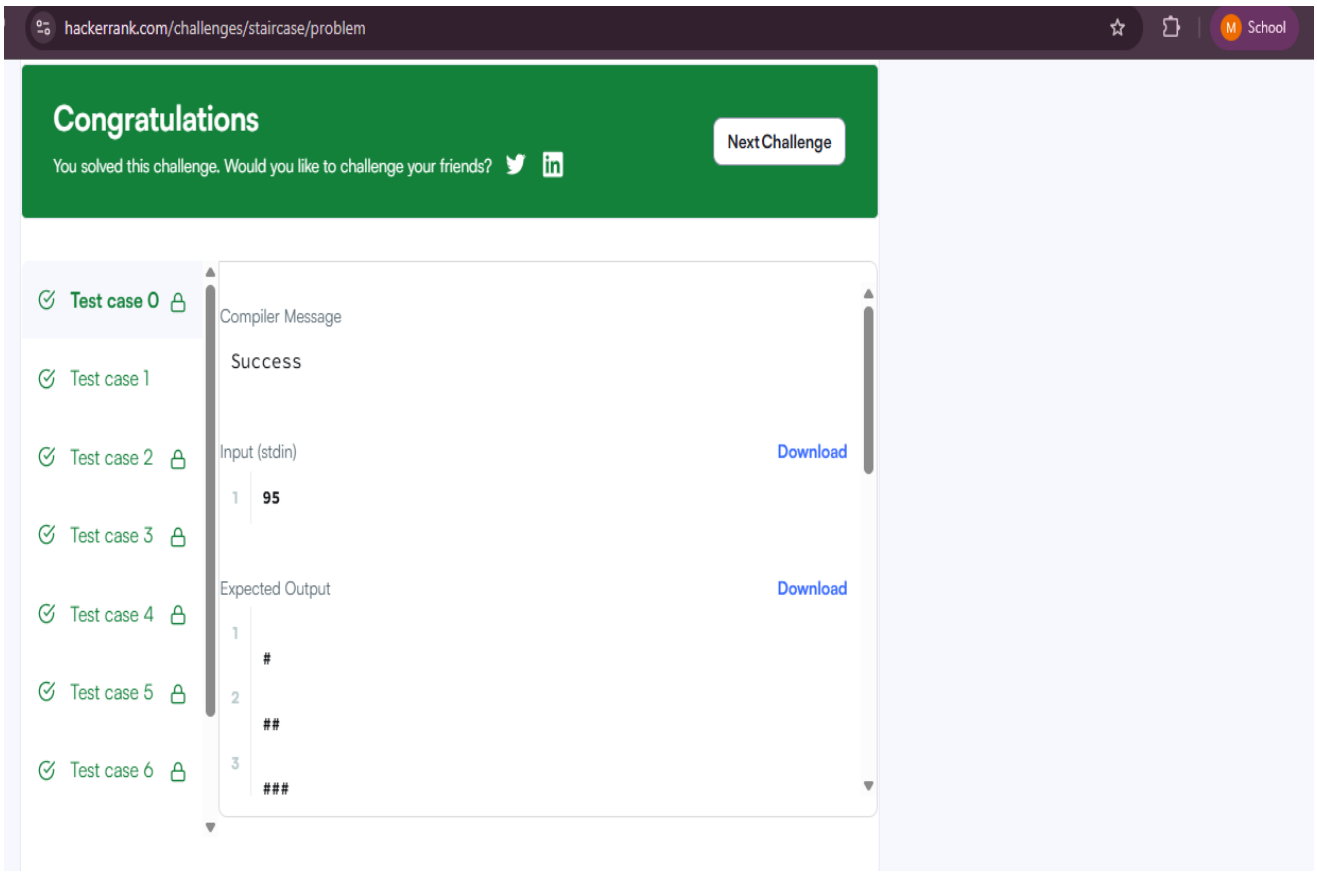
Module 6 — Nested loops: online judge problems

Question 29: Staircase

HackerRank

Answer:

```
void staircase(int n) {
    for (int i = 1; i <= n; i++) {
        for (int j = 0; j < n - i; j++) {
            printf(" ");
        }
        for (int j = 0; j < i; j++) {
            printf("#");
        }
        printf("\n");
    }
}
```



Question 30: Printing Pattern Using Loops

HackerRank

Answer:

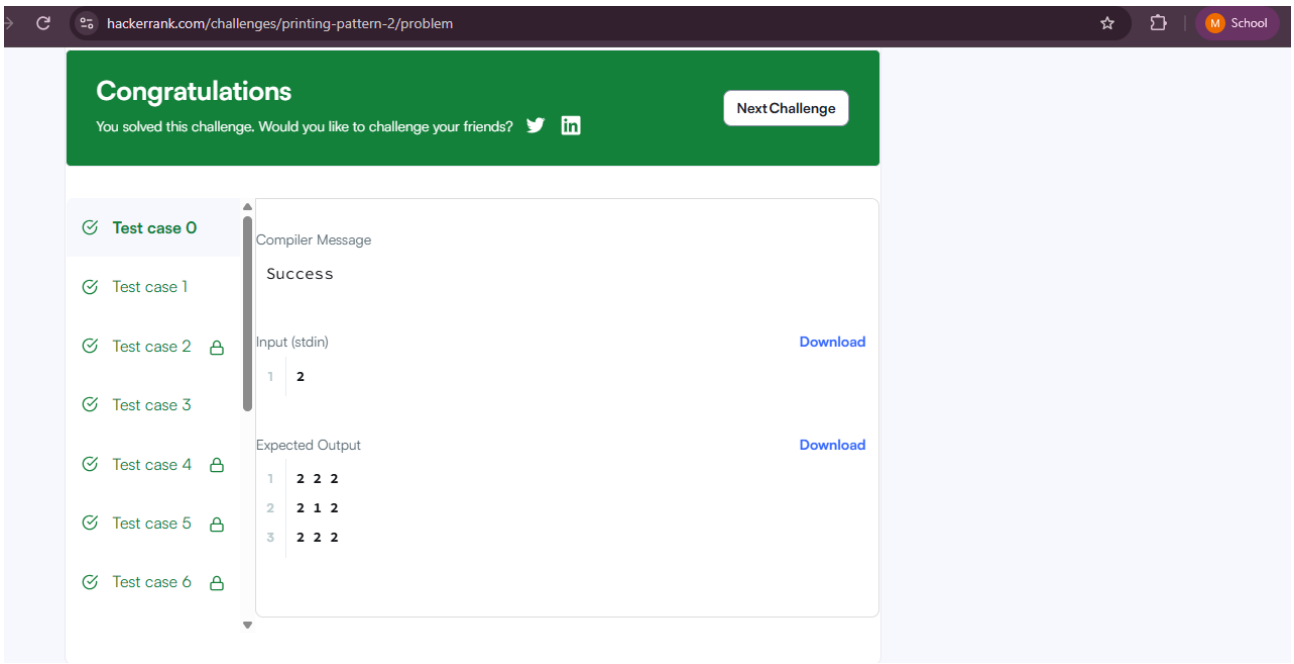
```
#include <stdio.h>
#include <string.h>
#include <math.h>
#include <stdlib.h>

int main()
{
    int n;
    scanf("%d", &n);

    int size = 2 * n - 1;

    for (int i = 0; i < size; i++) {
        for (int j = 0; j < size; j++) {
            int d = i;
            if (j < d) d = j;
            if (size - 1 - i < d) d = size - 1 - i;
            if (size - 1 - j < d) d = size - 1 - j;

            printf("%d", n - d);
            if (j < size - 1) printf(" ");
        }
        printf("\n");
    }
    return 0;
}
```



Module 7 — Nested-loop and pattern-printing questions

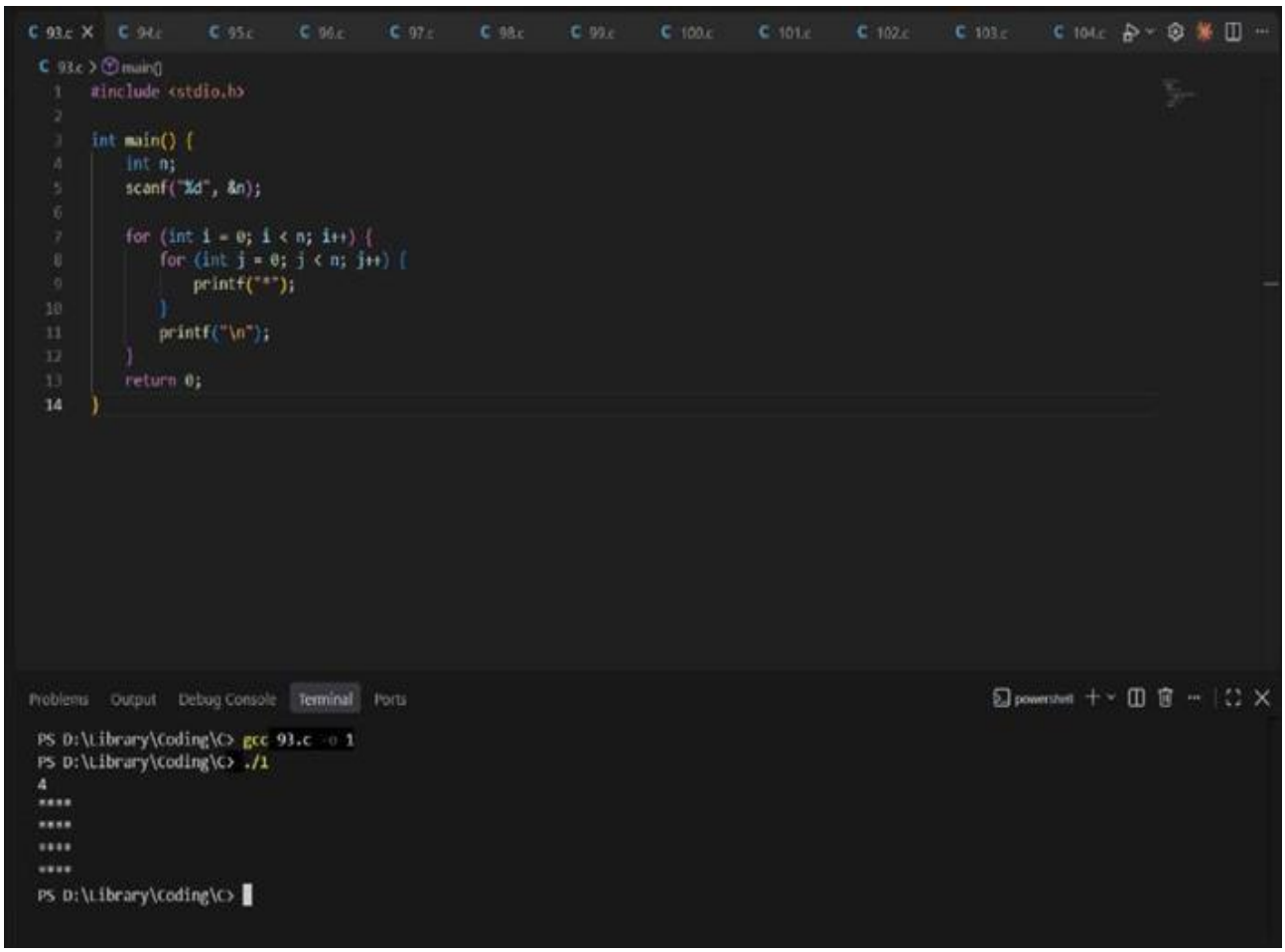
Question 31: Print a Solid Square

Answer:

```
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```



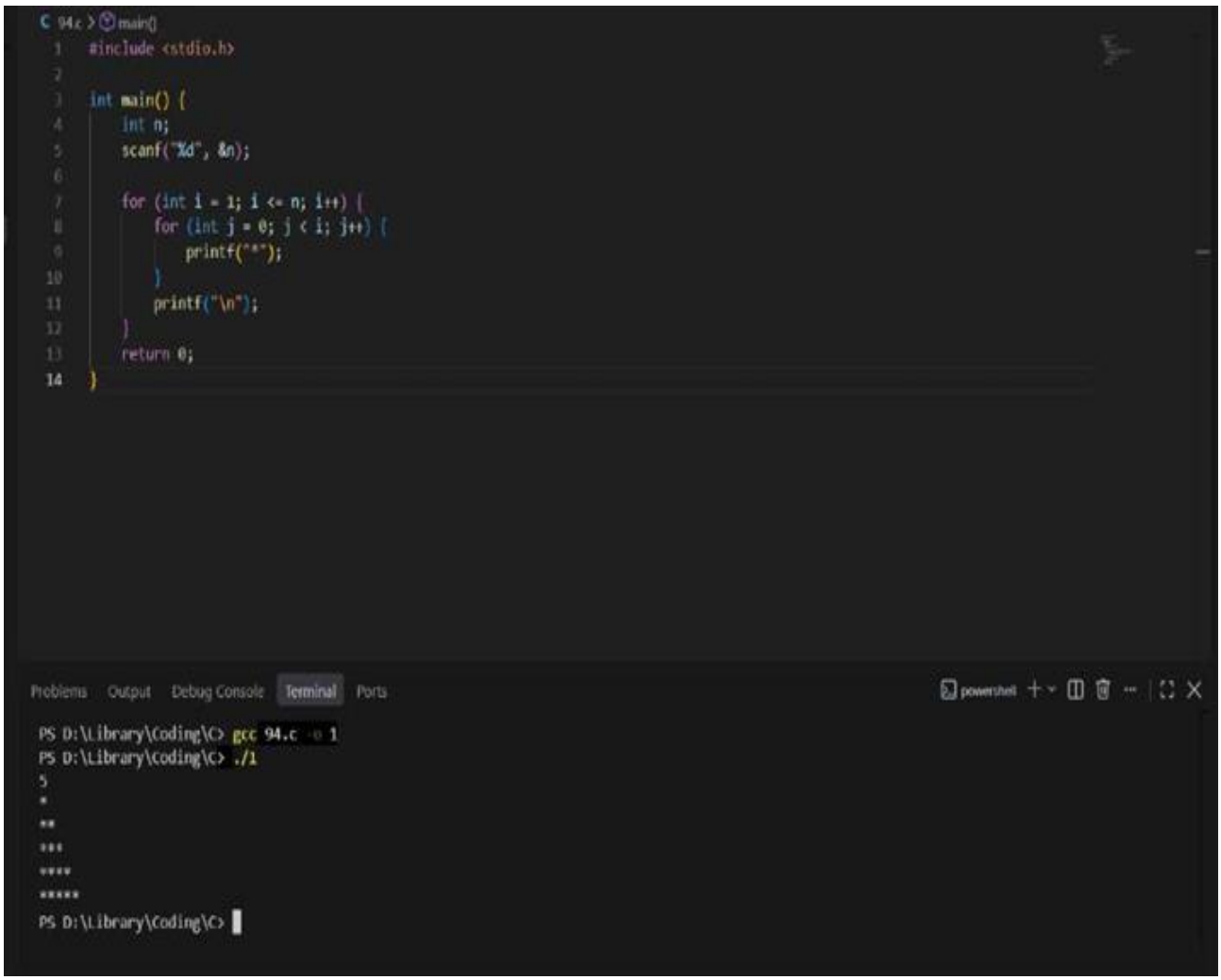
Question 32: Print a Right-Angled Triangle

Answer:

```
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    for (int i = 1; i <= n; i++) {
        for (int j = 0; j < i; j++) {
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```



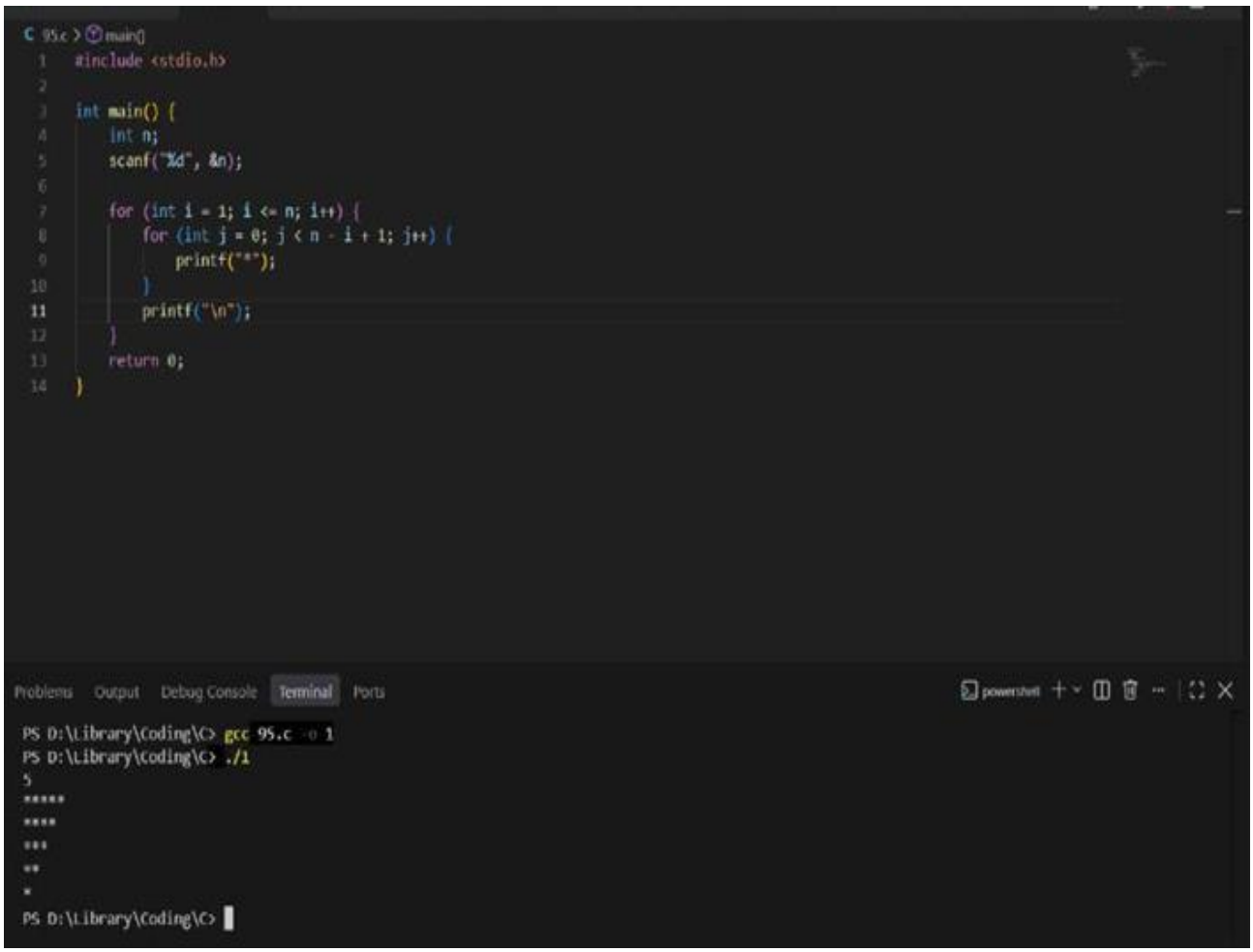
Question 33: Print an Inverted Triangle

Answer:

```
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    for (int i = 1; i <= n; i++) {
        for (int j = 0; j < n - i + 1; j++) {
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```



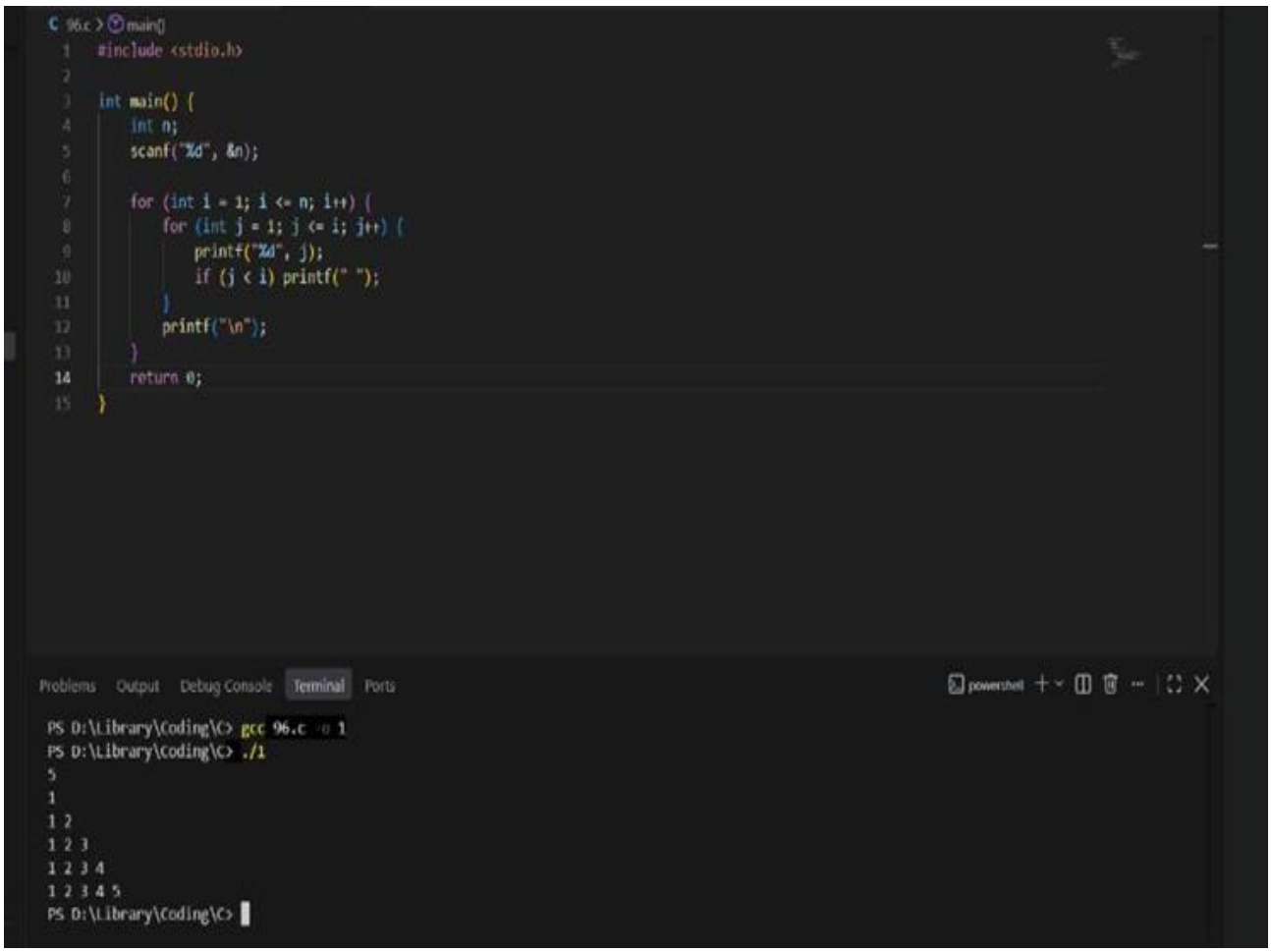
Question 34: Print a Number Triangle

Answer:

```
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) {
            printf("%d", j);
            if (j < i) printf(" ");
        }
        printf("\n");
    }
    return 0;
}
```



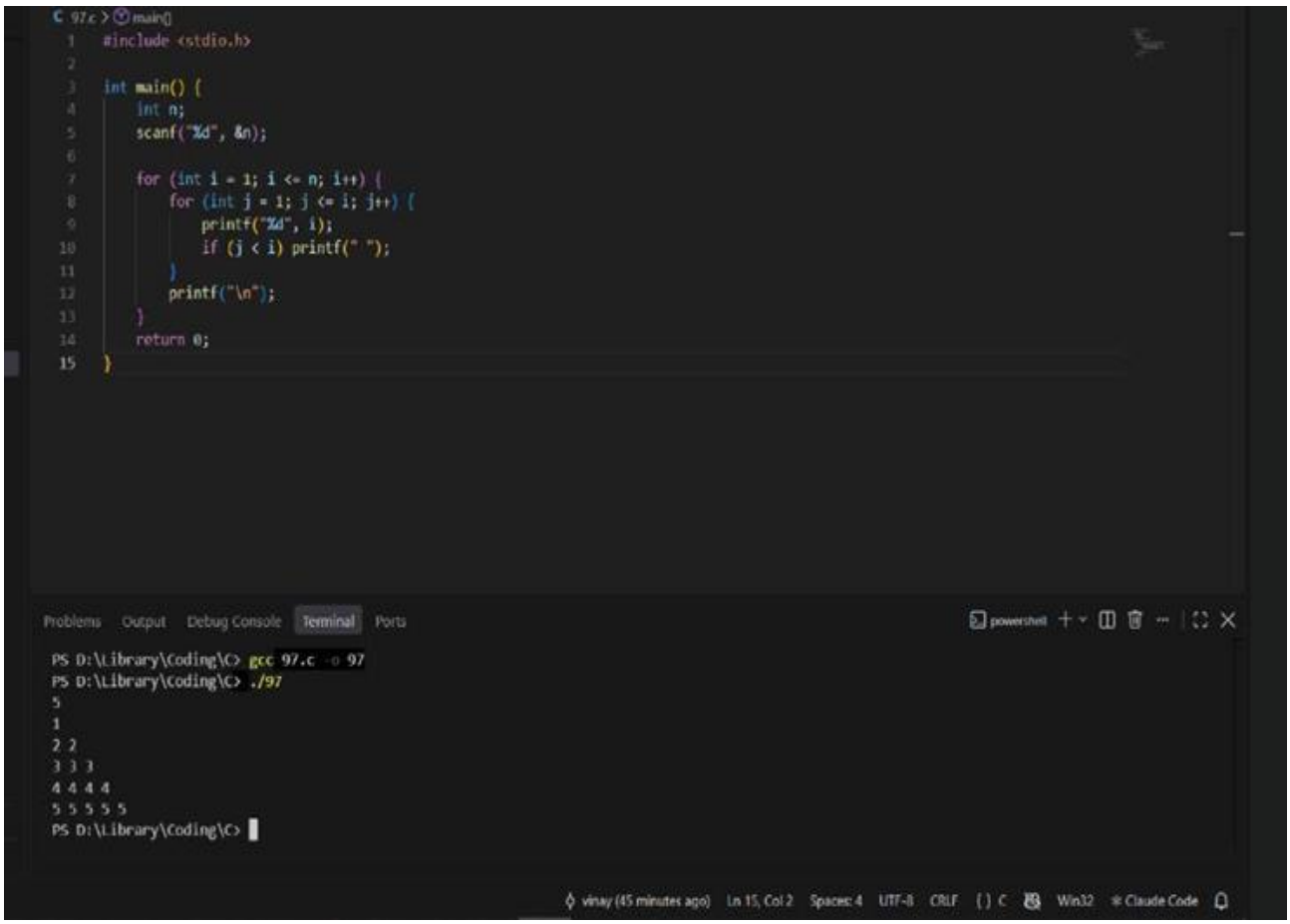
Question 35: Print a Repeated-Number Triangle

Answer:

```
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) {
            printf("%d", i);
            if (j < i) printf(" ");
        }
        printf("\n");
    }
    return 0;
}
```



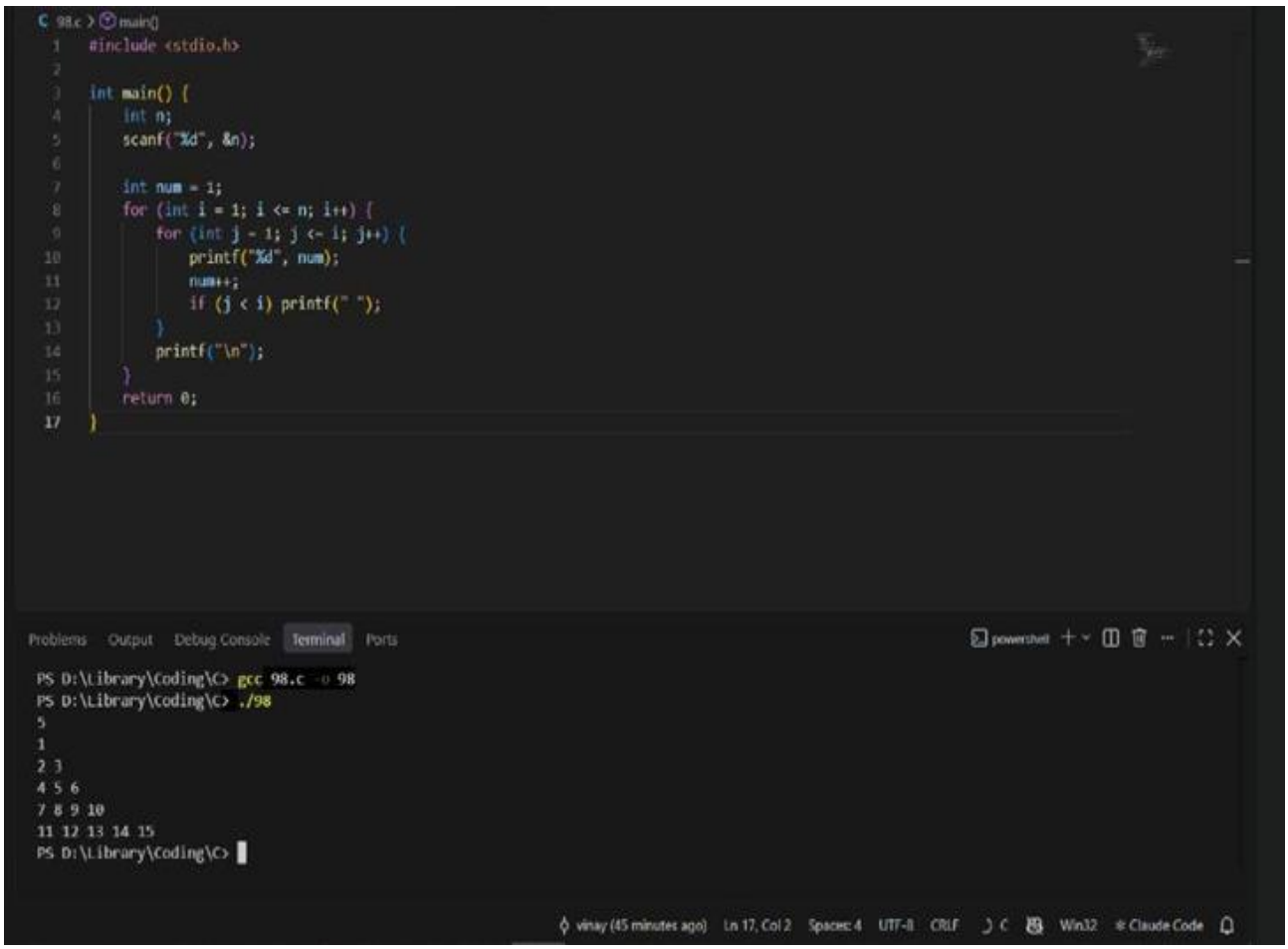
Question 36: Print a Consecutive-Number Triangle

Answer:

```
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int num = 1;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) {
            printf("%d", num);
            num++;
            if (j < i) printf(" ");
        }
        printf("\n");
    }
    return 0;
}
```



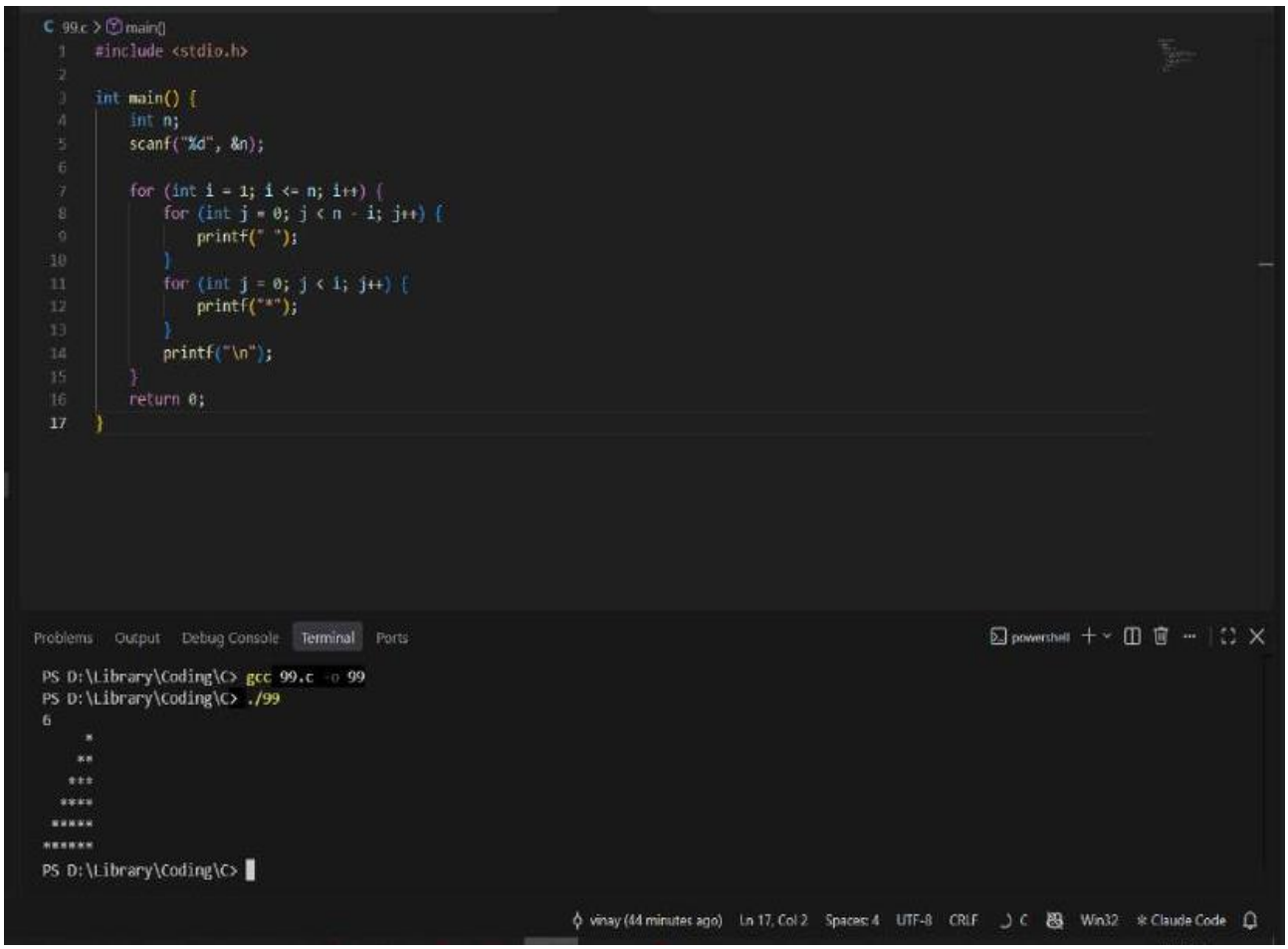
Question 37: Print a Right-Aligned Star Triangle

Answer:

```
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    for (int i = 1; i <= n; i++) {
        for (int j = 0; j < n - i; j++) {
            printf(" ");
        }
        for (int j = 0; j < i; j++) {
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```



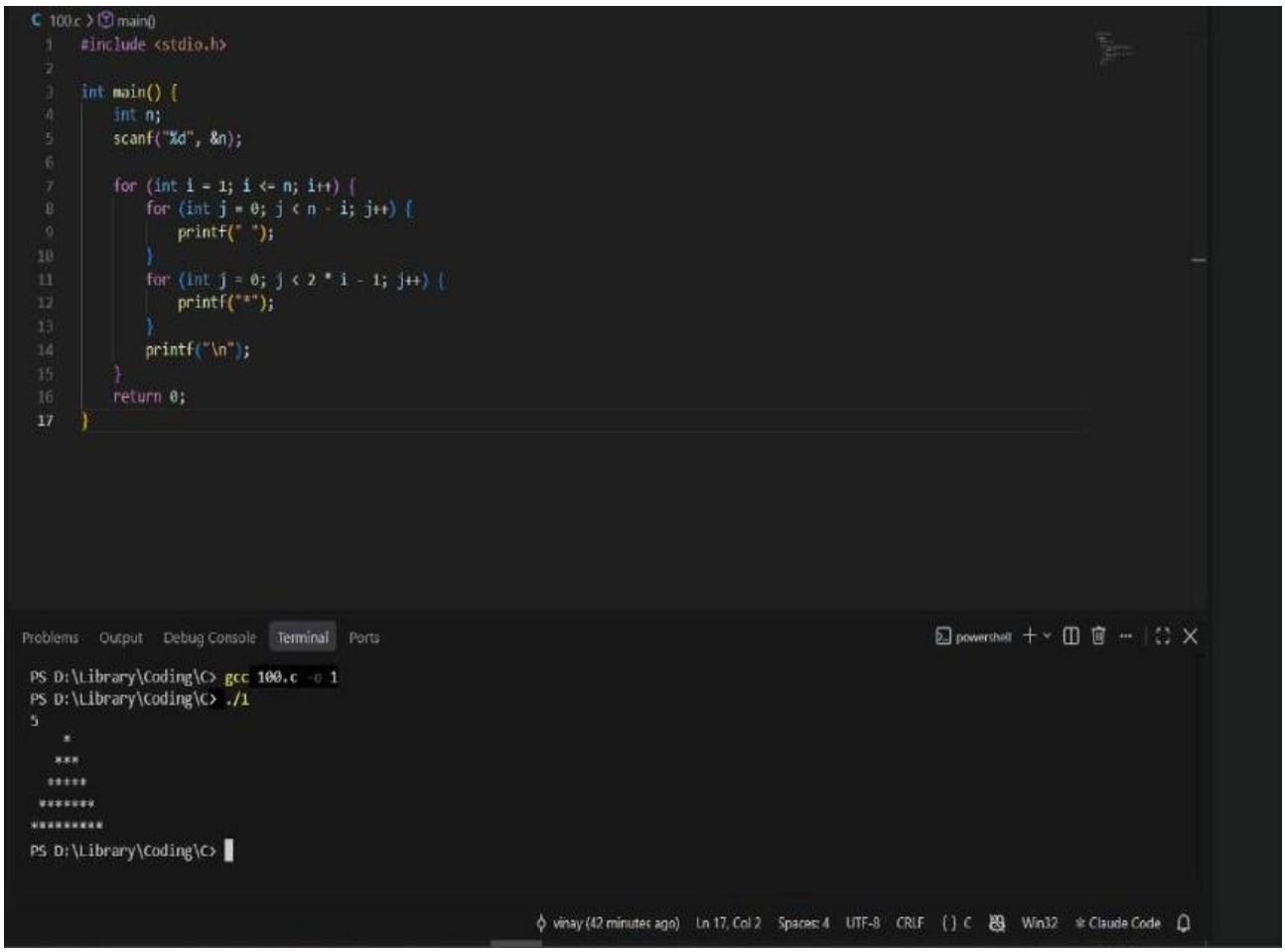
Question 38: Print a Centered Pyramid

Answer:

```
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    for (int i = 1; i <= n; i++) {
        for (int j = 0; j < n - i; j++) {
            printf(" ");
        }
        for (int j = 0; j < 2 * i - 1; j++) {
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```



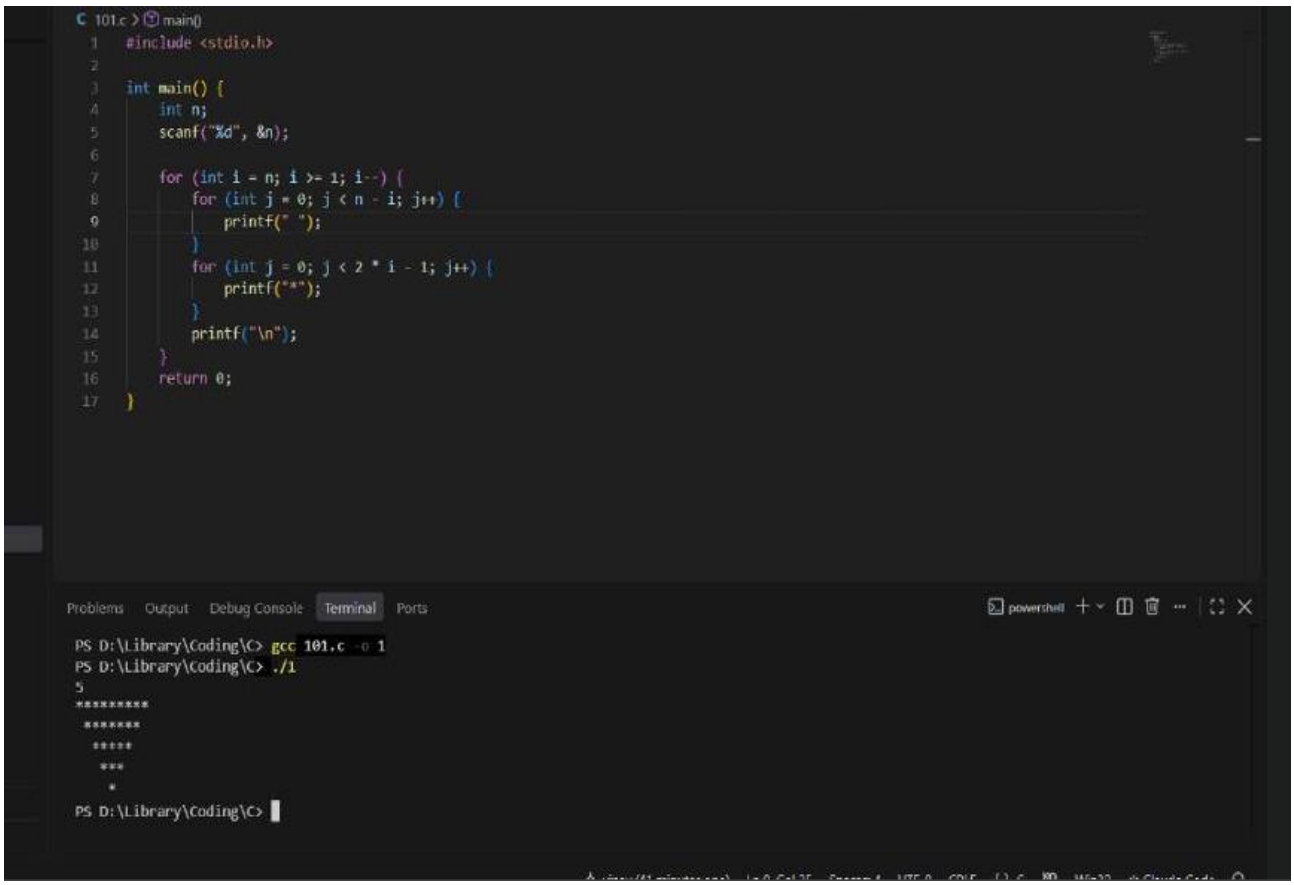
Question 39: Print an Inverted Centered Pyramid

Answer:

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    for (int i = n; i >= 1; i--) {
        for (int j = 0; j < n - i; j++) {
            printf(" ");
        }
        for (int j = 0; j < 2 * i - 1; j++) {
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```



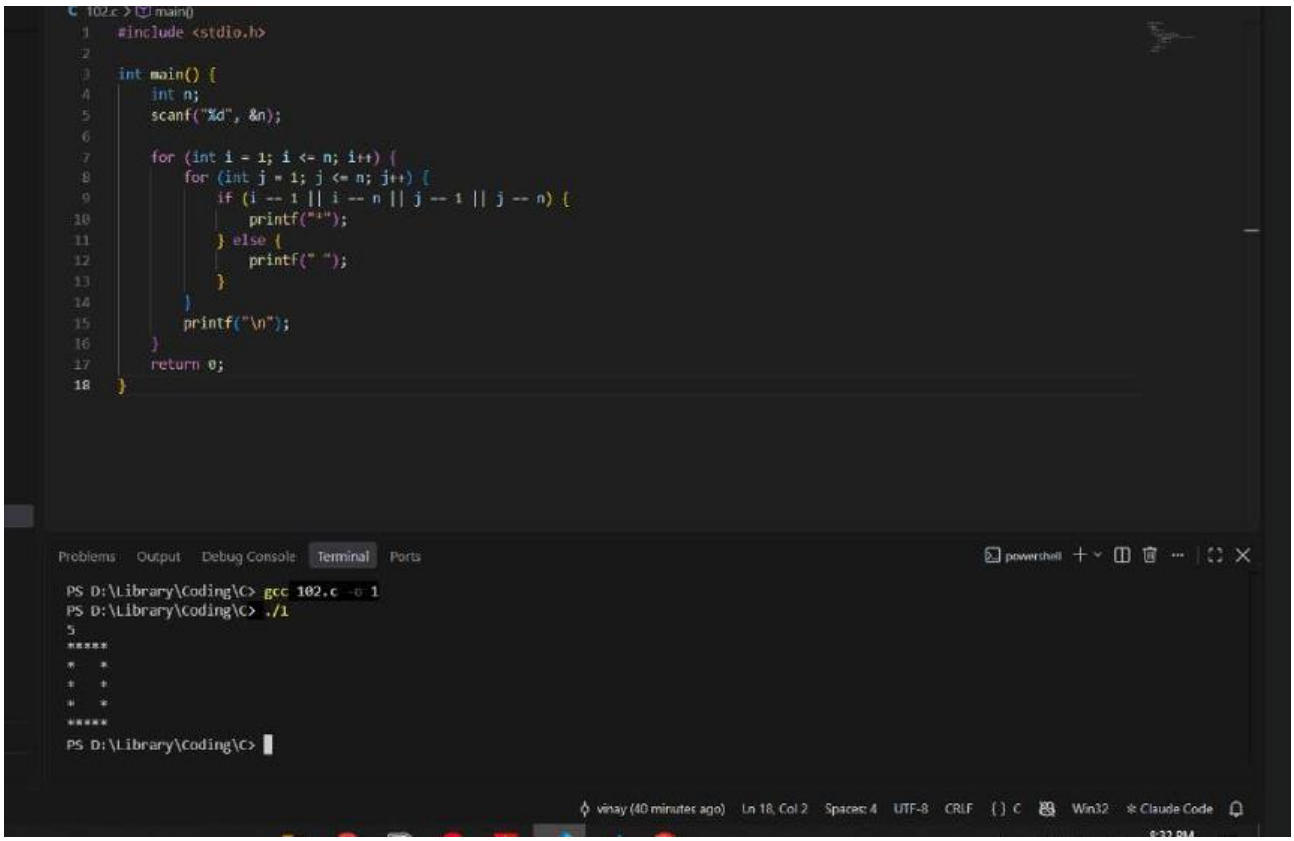
Question 40: Print a Hollow Square

Answer:

```
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {
            if (i == 1 || i == n || j == 1 || j == n) {
                printf("*");
            } else {
                printf(" ");
            }
        }
        printf("\n");
    }
    return 0;
}
```



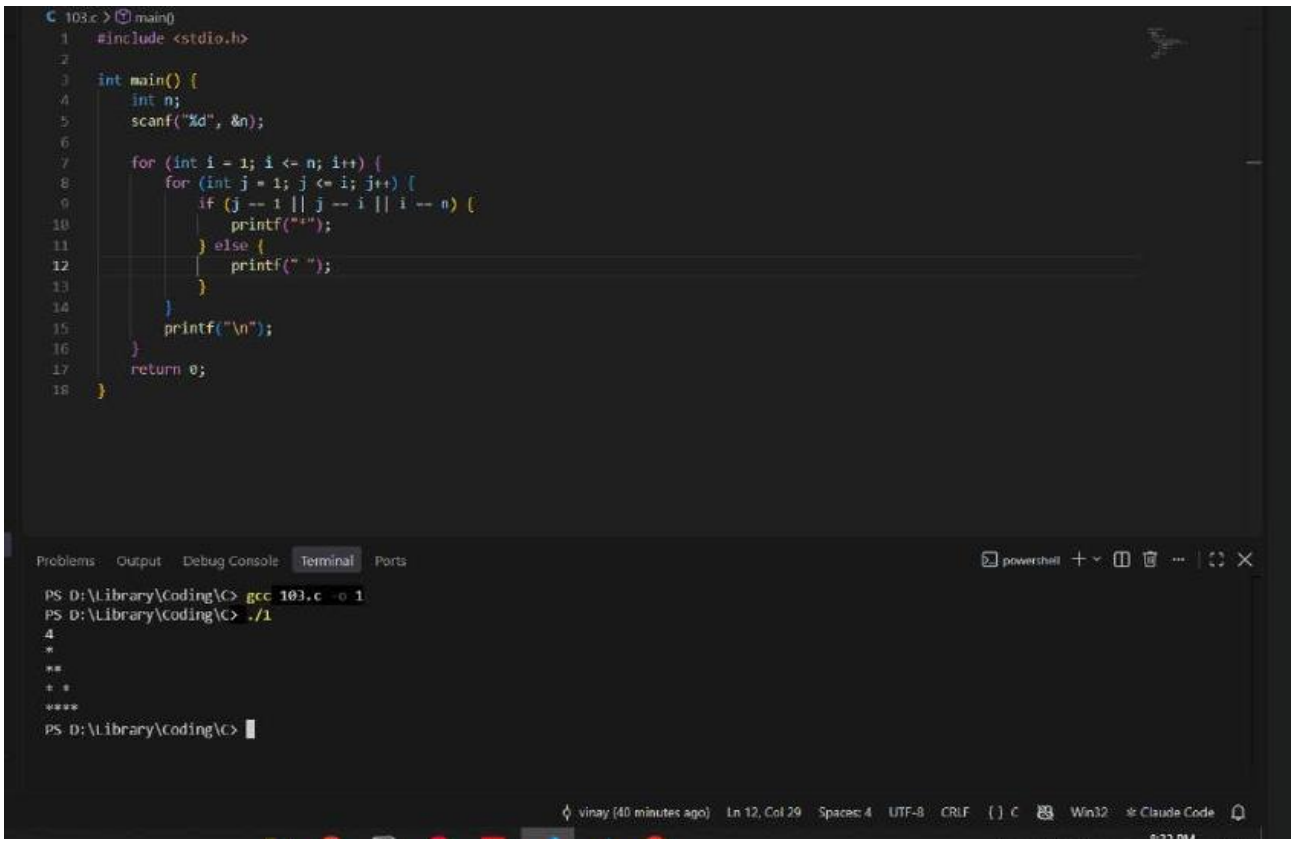
Question 41: Print a Hollow Right-Angled Triangle

Answer:

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) {
            if (j == 1 || j == i || i == n) {
                printf("*");
            } else {
                printf(" ");
            }
        }
        printf("\n");
    }
    return 0;
}
```



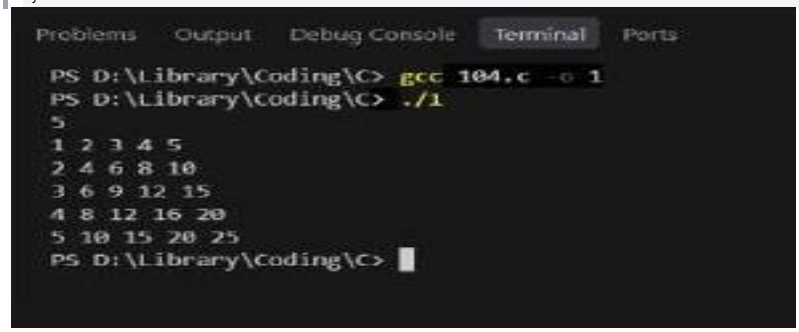
Question 42: Print a Multiplication Table Grid

Answer:

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {
            printf("%d", i * j);
            if (j < n) printf(" ");
        }
        printf("\n");
    }
    return 0;
}
```



Question 43: Print a Checkerboard

Answer:

```
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if ((i + j) % 2 == 0) {
                printf("X");
            } else {
                printf("O");
            }
        }
        printf("\n");
    }
    return 0;
}
```



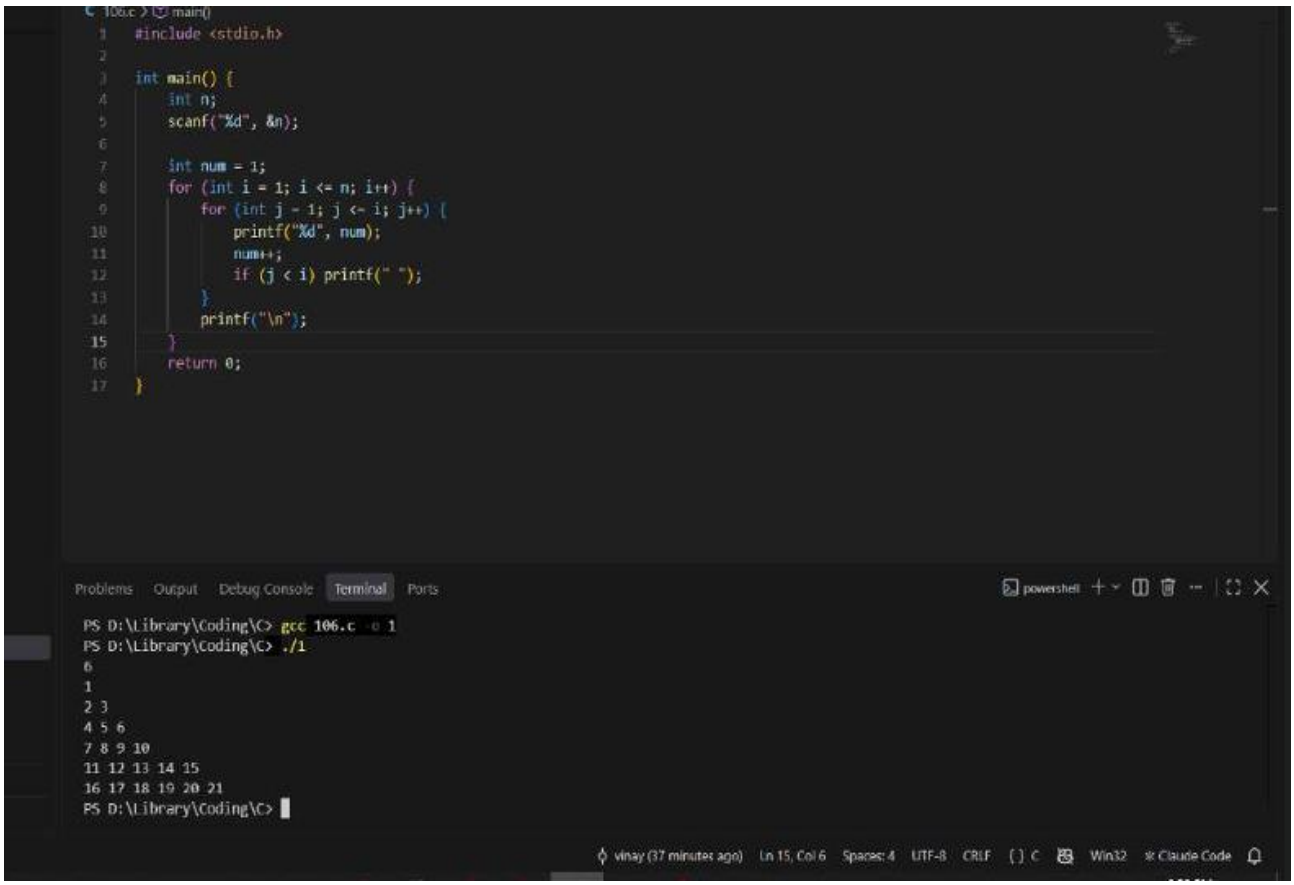
Question 44: Print Floyd's Triangle

Answer:

```
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int num = 1;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) {
            printf("%d", num);
            num++;
            if (j < i) printf(" ");
        }
        printf("\n");
    }
    return 0;
}
```



Question 45: Print a Butterfly Pattern

Answer:

```
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    // upper half
    for (int i = 1; i <= n; i++) {
        for (int j = 0; j < i; j++) printf("*");
        for (int j = 0; j < 2 * (n - i); j++) printf(" ");
        for (int j = 0; j < i; j++) printf("*");
        printf("\n");
    }

    // lower half (mirror)
    for (int i = n; i >= 1; i--) {
        for (int j = 0; j < i; j++) printf("*");
        for (int j = 0; j < 2 * (n - i); j++) printf(" ");
        for (int j = 0; j < i; j++) printf("*");
        printf("\n");
    }
    return 0;
}
```



Question 46: Print a Diamond Pattern

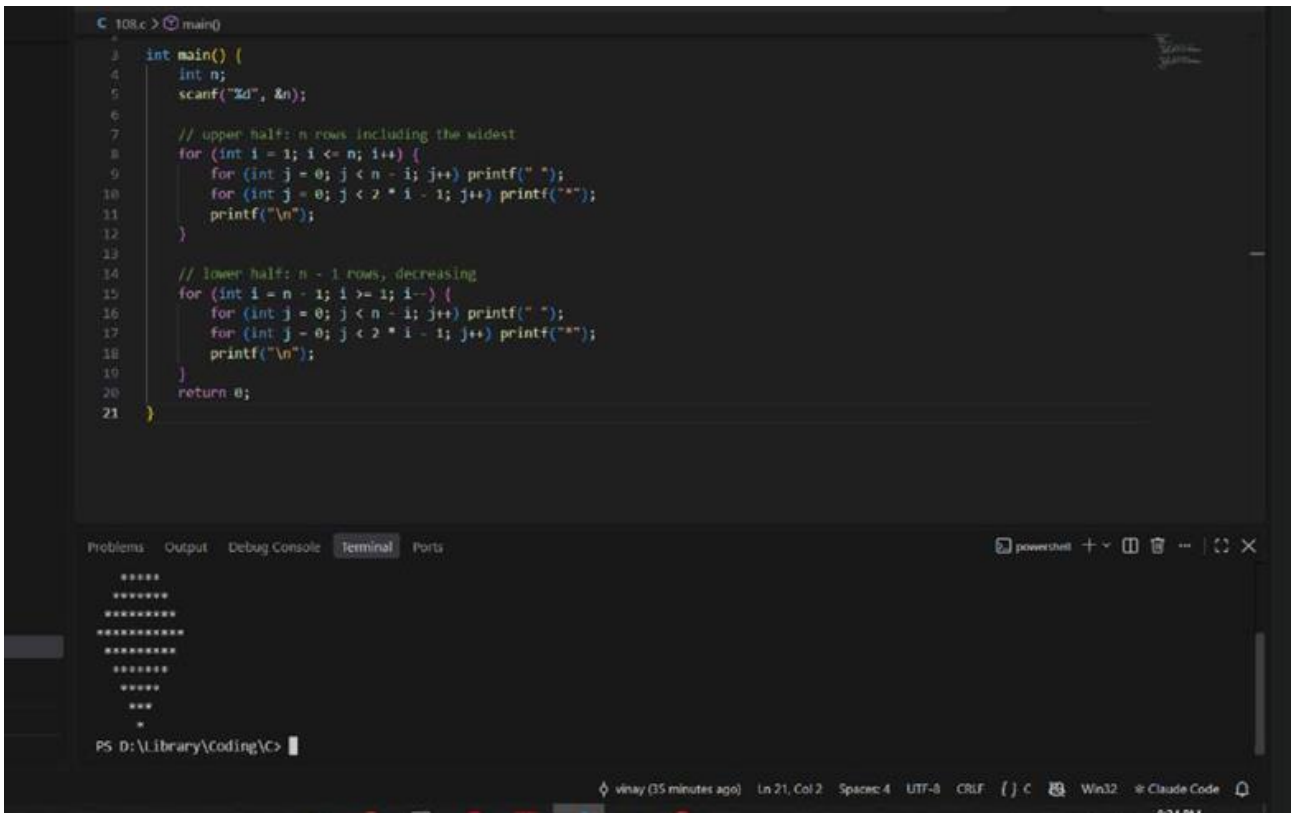
Answer:

```
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    // upper half: n rows including the widest
    for (int i = 1; i <= n; i++) {
        for (int j = 0; j < n - i; j++) printf(" ");
        for (int j = 0; j < 2 * i - 1; j++) printf("*");
        printf("\n");
    }

    // lower half: n - 1 rows, decreasing
    for (int i = n - 1; i >= 1; i--) {
        for (int j = 0; j < n - i; j++) printf(" ");
        for (int j = 0; j < 2 * i - 1; j++) printf("*");
        printf("\n");
    }
    return 0;
}
```


108.c > main()
int main() {
int n;
scanf("%d", &n);
// upper half: n rows including the widest
for (int i = 1; i <= n; i++) {
for (int j = 0; j < n - i; j++) printf(" ");
for (int j = 0; j < 2 * i - 1; j++) printf("*");
printf("\n");
}
// lower half: n - 1 rows, decreasing
for (int i = n - 1; i >= 1; i--) {
for (int j = 0; j < n - i; j++) printf(" ");
for (int j = 0; j < 2 * i - 1; j++) printf("*");
printf("\n");
}
return 0;
}
Problems
Output
Debug Console
Terminal
Ports
powershell
*****
****
****
****
****
****
*****
***
*
PS D:\Library\Coding\C>
vinay (35 minutes ago)
Ln 21, Col 2
Spaces: 4
UTF-8
CRLF
{} C
Win32
Claude Code